💫World Air Quality Index | EDA 🌤️

Data Analytics, EDA

AQI.png

1. Introduction

In this notebook, we will perform Exploratory Data Analysis (EDA) on the World Air Quality Index dataset. The dataset contains information about air quality measurements in different cities around the world. We will explore the data, visualize key insights, and derive meaningful analytics from it.

About Dataset

Sulfur Dioxide (SO2): SO2 is a gas produced by the burning of fossil fuels, particularly in industrial processes. It can have harmful effects on human health, leading to respiratory issues and exacerbating asthma and other respiratory conditions.

Nitrogen Dioxide (NO2): NO2 is a gas produced by the burning of fossil fuels, primarily in vehicles and power plants. Exposure to high levels of NO2 can irritate the respiratory system and contribute to the development of respiratory conditions.

Air Quality Index (AQI): AQI is a measure used to communicate the level of air pollution in a specific area. It is often calculated based on the concentrations of various pollutants, such as PM2.5, ozone, CO, SO2, and NO2. The AQI provides a standardized scale to help individuals understand the air quality and take necessary precautions to protect their health.

Good: The “Good” category on the AQI scale indicates that the air quality is satisfactory, and there is little to no health risk associated with it.

Moderate: The “Moderate” category signifies that the air quality is acceptable, but there may be a moderate health concern for a small number of individuals, particularly those who are unusually sensitive to air pollution.

Unhealthy for Sensitive Groups: This category indicates that the air quality is of concern for individuals who are more vulnerable to the effects of air pollution, such as children, older adults, and people with respiratory or cardiovascular conditions.

Unhealthy: The “Unhealthy” category suggests that the general population may start to experience health effects due to the poor air quality. It is advisable to limit outdoor activities and take necessary precautions.

Very Unhealthy: The “Very Unhealthy” category signifies a significant health risk from the air pollution. It is recommended to avoid outdoor activities and minimize exposure to the polluted air.

Hazardous: The “Hazardous” category indicates the highest level of air pollution, posing a severe health risk. It is crucial to stay indoors, use air

About Dataset

filtration systems, and follow local authorities' guidance to protect health.

2. Dataset Overview

Let's start by loading the dataset and getting a brief overview of its contents.

```
# Import necessary libraries
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

import pandas as pd

# Load the dataset
california_data = pd.read_csv("/content/sample_data/california_housing_train.c

# Display the first few rows of the dataset
california_data.head()
```

	longitude	latitude	housing_median_age	total_rooms	total_bedrooms	populati
0	-114.31	34.19	15.0	5612.0	1283.0	1015.0
1	-114.47	34.40	19.0	7650.0	1901.0	1129.0
2	-114.56	33.69	17.0	720.0	174.0	333.0
3	-114.57	33.64	14.0	1501.0	337.0	515.0
4	-114.57	33.57	20.0	1454.0	326.0	624.0

```
<div class="colab-df-buttons">


<button class="colab-df-convert" onclick="convertToInteractive('df-695d0967-5c
        title="Convert this dataframe to an interactive table."
        style="display:none;">
```

```
</div>
```

3. Data Cleaning and Preprocessing

Before diving into the analysis, it’s important to clean and preprocess the data to ensure accurate and meaningful results. We will handle missing values, remove irrelevant columns, and perform any necessary transformations.

```
california_data.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 17000 entries, 0 to 16999
Data columns (total 9 columns):
 #   Column              Non-Null Count  Dtype
---  ------              --------------  -----
 0   longitude           17000 non-null  float64
 1   latitude            17000 non-null  float64
 2   housing_median_age  17000 non-null  float64
 3   total_rooms         17000 non-null  float64
 4   total_bedrooms      17000 non-null  float64
 5   population          17000 non-null  float64
 6   households          17000 non-null  float64
 7   median_income       17000 non-null  float64
 8   median_house_value  17000 non-null  float64
dtypes: float64(9)
memory usage: 1.2 MB
```

Dataset contains **16,695 rows** and ****14 columns.** Here is a summary of the columns and their data types:

#	Column	Dtype
0	Country	object
1	City	object
2	AQI Value	int64
3	AQI Category	object

#	Column	Dtype
4	CO AQI Value	int64
5	CO AQI Category	object
6	Ozone AQI Value	int64
7	Ozone AQI Category	object
8	NO2 AQI Value	int64
9	NO2 AQI Category	object
10	PM2.5 AQI Value	int64
11	PM2.5 AQI Category	object
12	lat	float64
13	lng	float64

```
california_data.columns
```

```
Index(['longitude', 'latitude', 'housing_median_age', 'total_rooms',
       'total_bedrooms', 'population', 'households', 'median_income',
       'median_house_value'],
      dtype='object')
```

```
california_data.describe(include ='all')
```

	longitude	latitude	housing_median_age	total_rooms	total_bed
count	17000.000000	17000.000000	17000.000000	17000.000000	17000.000
mean	-119.562108	35.625225	28.589353	2643.664412	539.41082
std	2.005166	2.137340	12.586937	2179.947071	421.49945
min	-124.350000	32.540000	1.000000	2.000000	1.000000
25%	-121.790000	33.930000	18.000000	1462.000000	297.00000
50%	-118.490000	34.250000	29.000000	2127.000000	434.00000
75%	-118.000000	37.720000	37.000000	3151.250000	648.25000
max	-114.310000	41.950000	52.000000	37937.000000	6445.0000

```
<div class="colab-df-buttons">


<button class="colab-df-convert" onclick="convertToInteractive('df-b6e1ca0d-3b
        title="Convert this dataframe to an interactive table."
        style="display:none;">
```

```
</div>

print(california_data.columns)

Index(['longitude', 'latitude', 'housing_median_age', 'total_rooms',
       'total_bedrooms', 'population', 'households', 'median_income',
       'median_house_value'],
      dtype='object')
```

Based on the provided summary statistics of the dataset, we can derive the following insights:

1. **Air Quality Index (AQI) Value:** The average AQI value is approximately 63, with a minimum of 7 and a maximum of 500. This indicates a wide range of air quality conditions across the cities in the dataset.

2. **Pollutant Categories:** The dataset includes AQI values and categories for various pollutants such as **CO, Ozone, NO2, and PM2.5.** The most frequently recorded pollutant category is **"Good"** for all pollutants, suggesting that a majority of the cities have relatively satisfactory air quality.

3. **Geographical Distribution:** The dataset covers cities from 174 different countries. The most frequently represented country is the **United States of America**, with 3,954 occurrences. This indicates that the dataset is heavily skewed towards cities in the United States.

4. **Geographic Coordinates:** The dataset includes latitude and longitude coordinates for each city, providing spatial information for further analysis and visualization.

```
california_data.isnull().sum()
```

	0
longitude	0
latitude	0
housing_median_age	0
total_rooms	0
total_bedrooms	0
population	0
households	0
median_income	0
median_house_value	0

dtype: int64

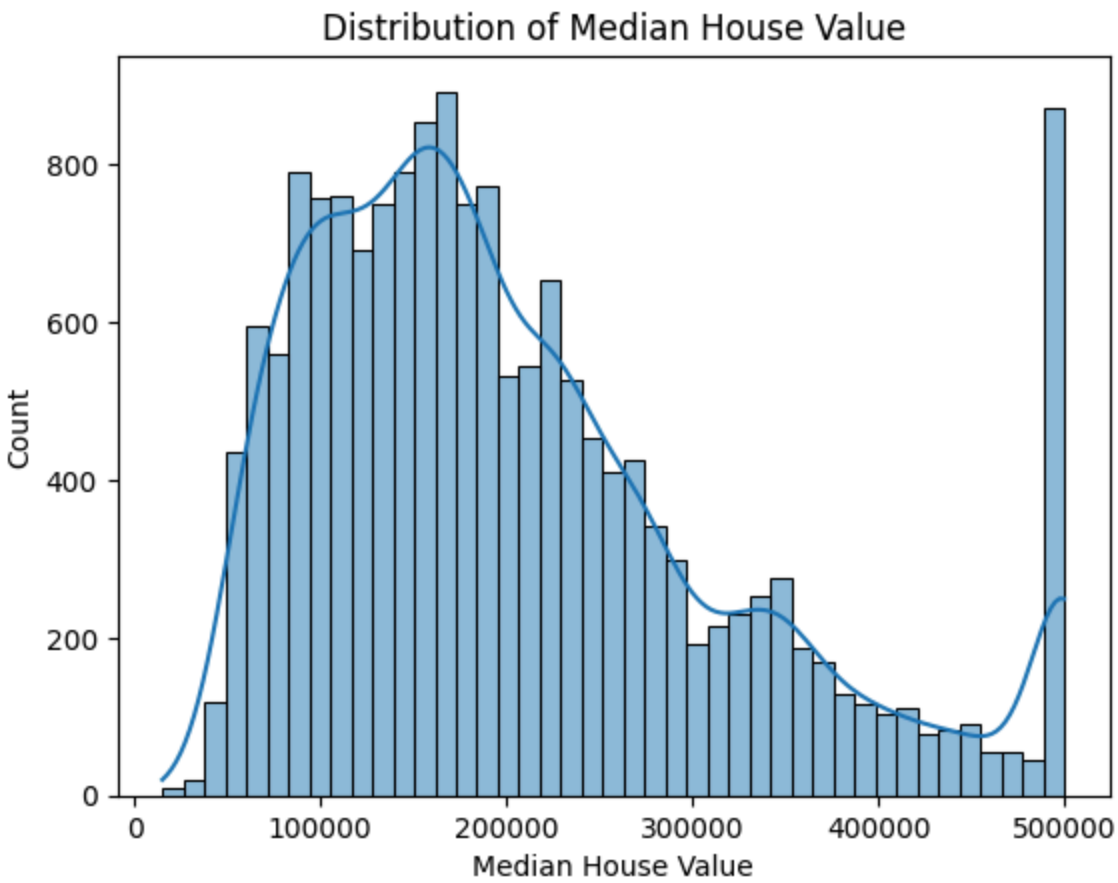
```
california_data.dropna(axis=1, inplace=True)
```

4. Exploratory Data Analysis

Now, let’s explore the dataset to gain insights into the air quality measurements. We will analyze various aspects such as pollutant levels, geographical distribution, and temporal patterns.

Distribution of the AQI Values

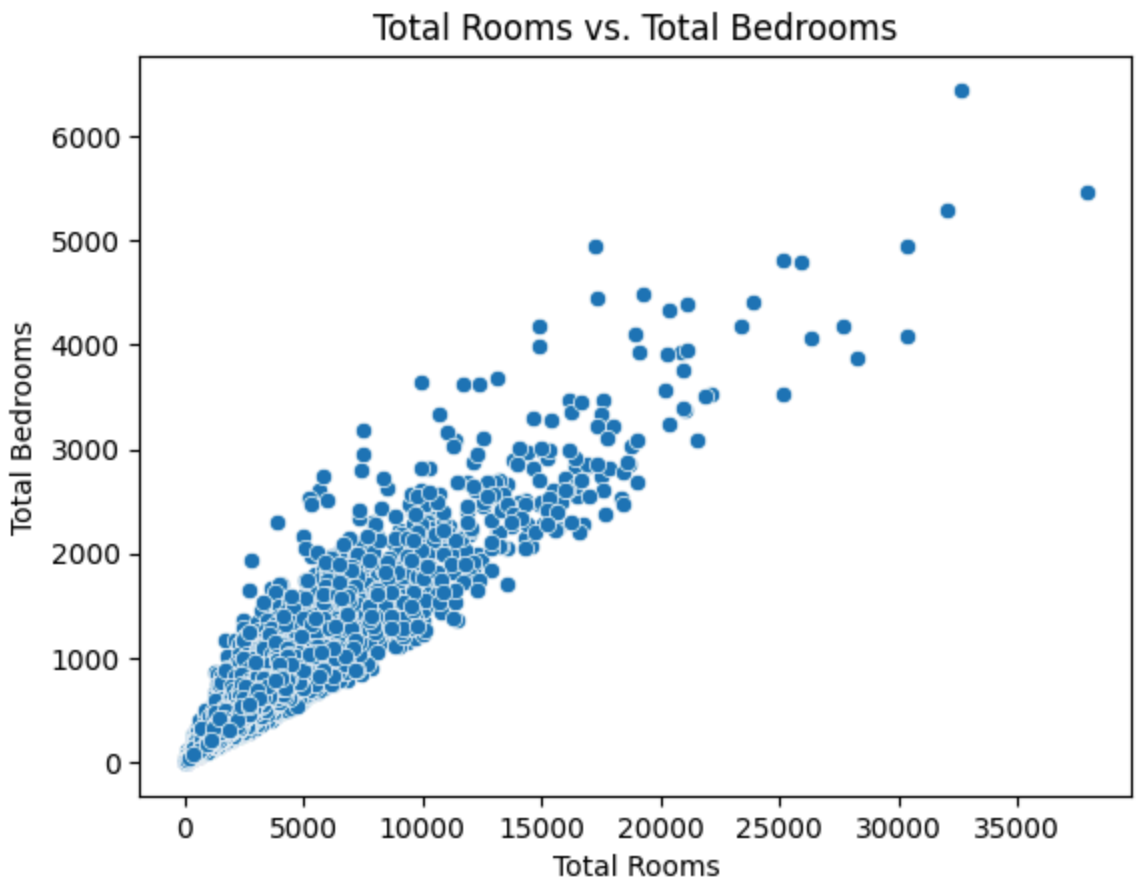
```
sns.histplot(california_data['median_house_value'], kde=True)
plt.title('Distribution of Median House Value')
plt.xlabel('Median House Value')
plt.ylabel('Count')
plt.show()
```



png

Relationship between AQI Value and other variables

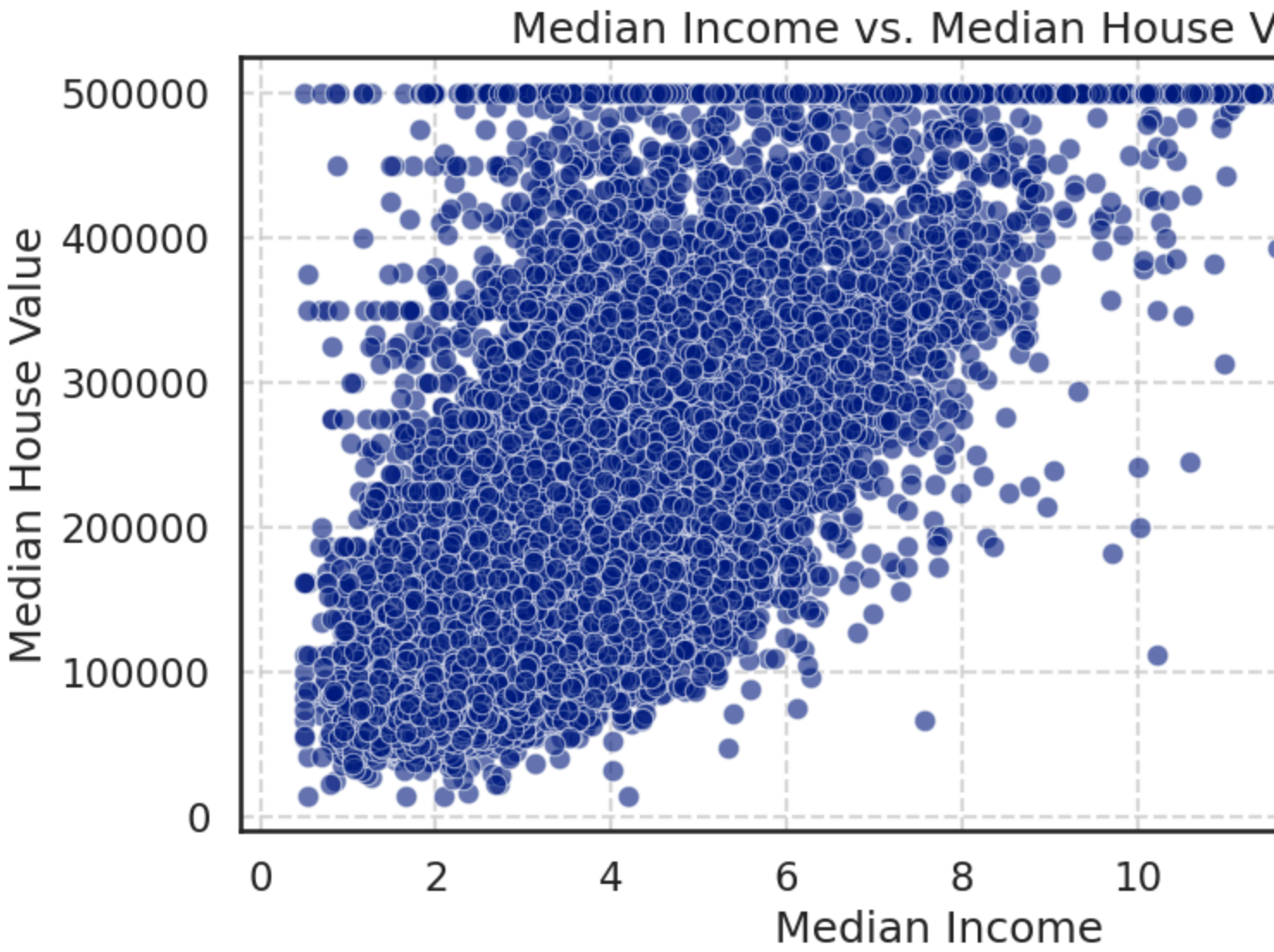
```
sns.scatterplot(data=california_data, x='total_rooms', y='total_bedrooms')
plt.title('Total Rooms vs. Total Bedrooms')
plt.xlabel('Total Rooms')
plt.ylabel('Total Bedrooms')
plt.show()
```



png

Relationship between Median Income and Median House Value

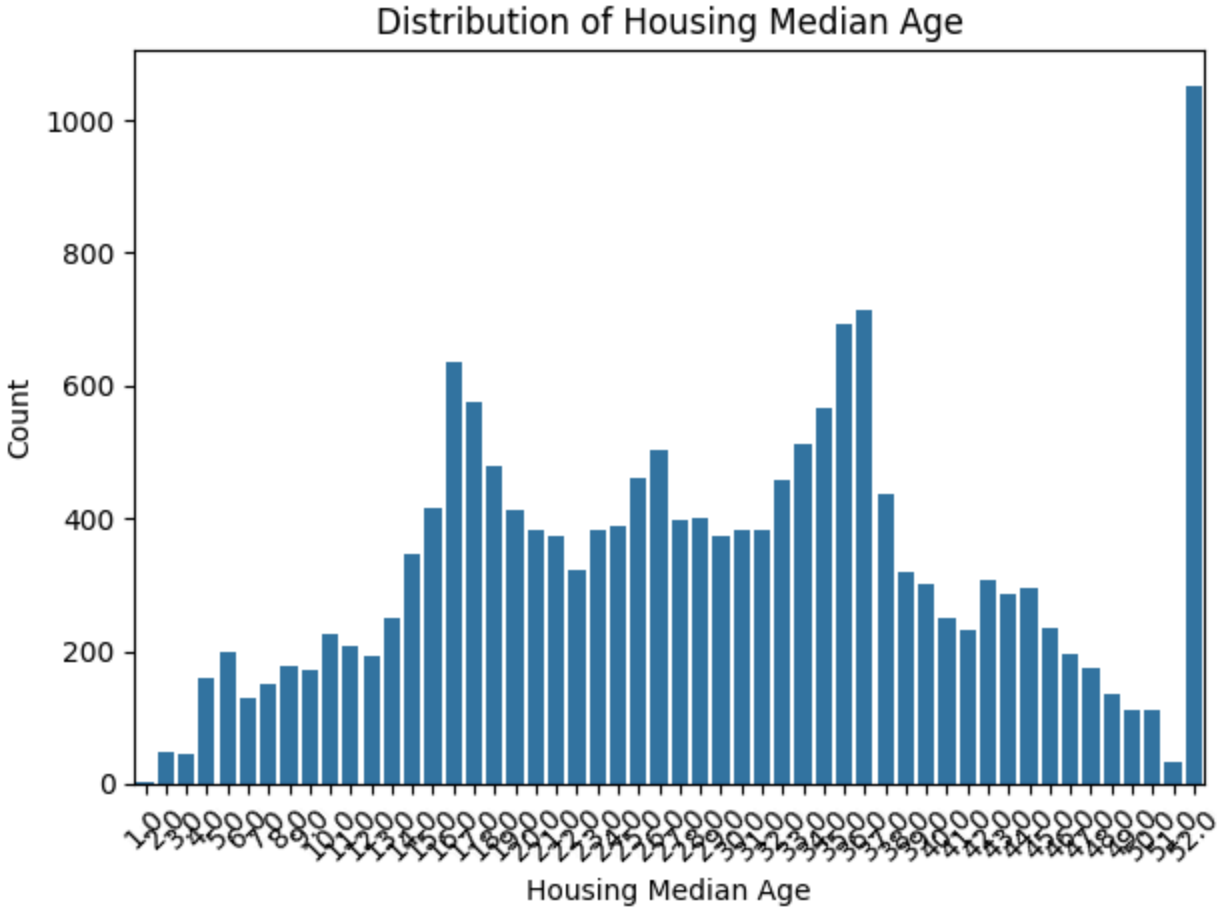
```
plt.figure(figsize=(10, 6))
sns.scatterplot(data=california_data, x='median_income', y='median_house_valu
plt.title('Median Income vs. Median House Value')
plt.xlabel('Median Income')
plt.ylabel('Median House Value')
plt.grid(True, linestyle='--', alpha=0.7)
plt.tight_layout()
plt.show()
```



png

Distribution of AQI categories

```
sns.countplot(data=california_data, x='housing_median_age')
plt.title('Distribution of Housing Median Age')
plt.xlabel('Housing Median Age')
plt.ylabel('Count')
plt.xticks(rotation=45)  # Rotate x-axis labels by 45 degrees
plt.tight_layout()  # Adjust the plot layout to prevent overlapping
plt.show()
```

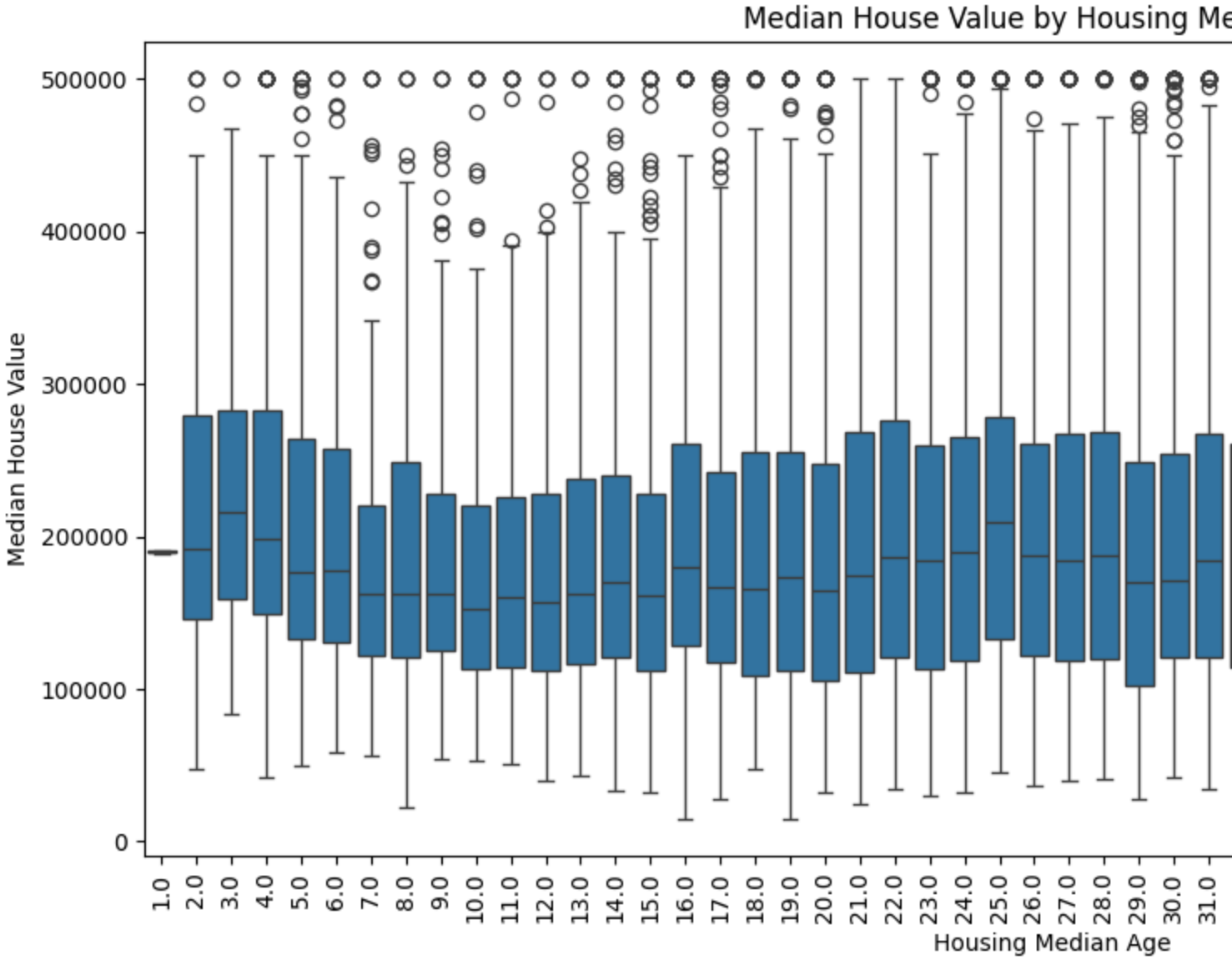


png

The countplot of AQI Categories by Country provides an overview of the distribution of air quality across different countries. It allows us to see the frequency of each AQI category (Good, Moderate, Unhealthy for Sensitive Groups, etc.) within each country. This information can help us identify countries with higher or lower air quality levels.

AQI Category vs. AQI Value

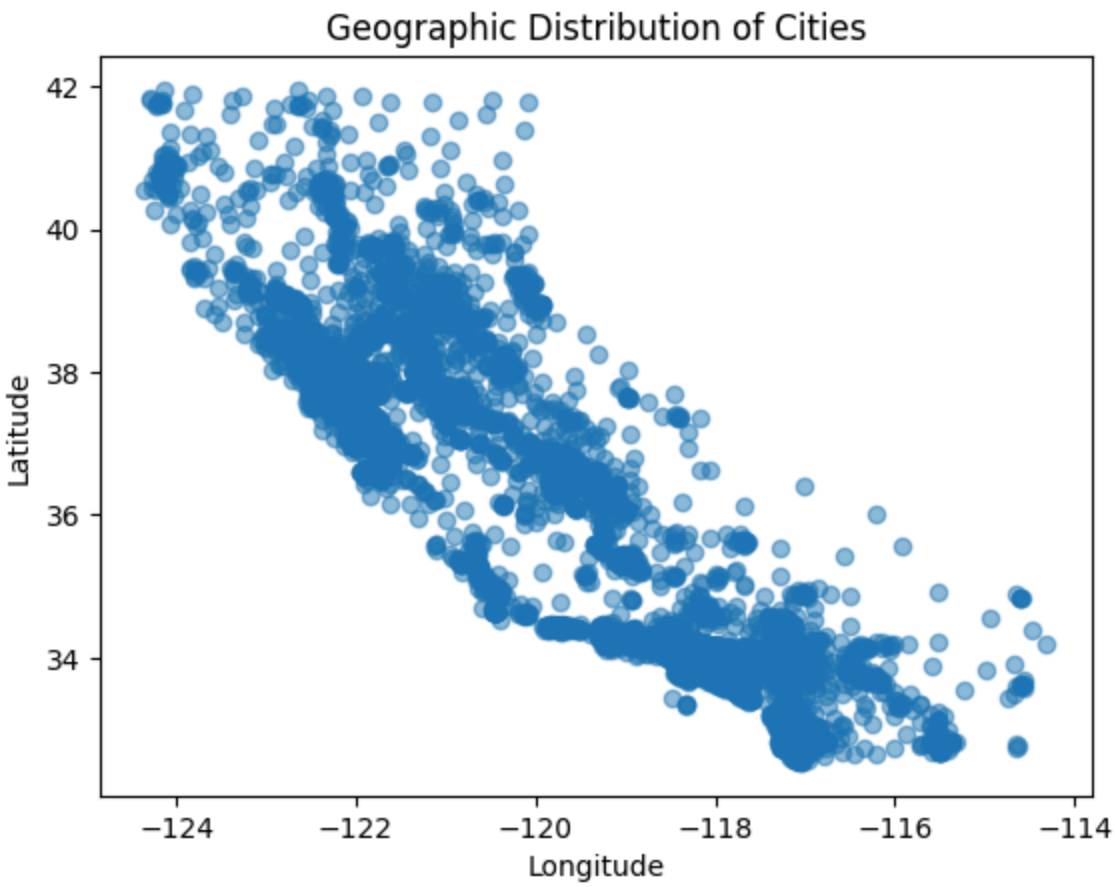
```
plt.figure(figsize=(12, 6))
sns.boxplot(data=california_data, x='housing_median_age', y='median_house_valu
plt.title('Median House Value by Housing Median Age')
plt.xlabel('Housing Median Age')
plt.ylabel('Median House Value')
plt.xticks(rotation=90)
plt.tight_layout()
plt.show()
```



png

Geographic distribution of cities

```
plt.scatter(california_data['longitude'], california_data['latitude'], alpha=0
plt.title('Geographic Distribution of Cities')
plt.xlabel('Longitude')
plt.ylabel('Latitude')
plt.show()
```

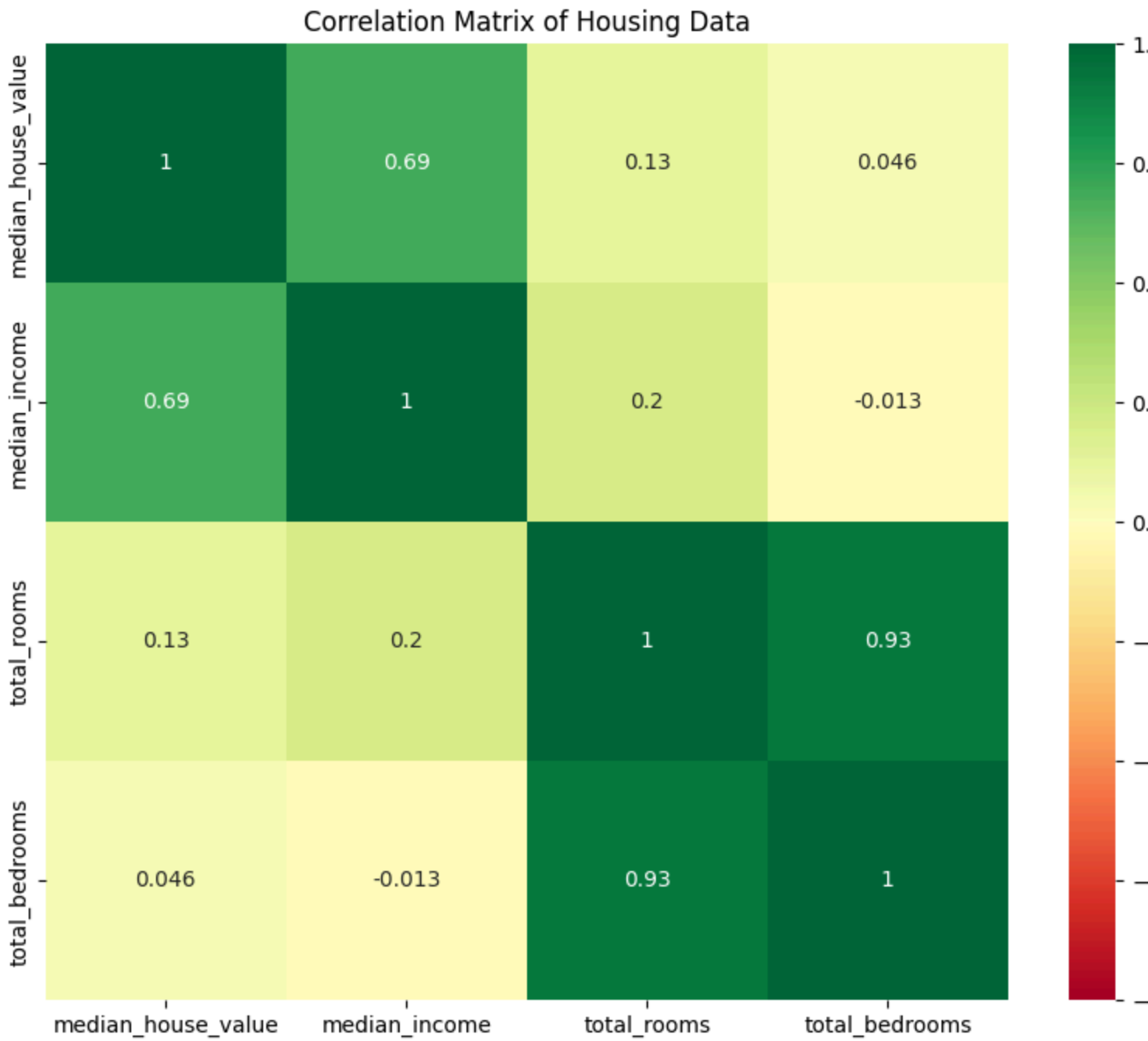


png

Comparison of AQI Categories by Pollutant

Correlation Matrix of Pollutant AQI Values

```
corr_matrix = california_data[['median_house_value', 'median_income', 'total_

plt.figure(figsize=(10, 8))
sns.heatmap(corr_matrix, annot=True, cmap='RdYlGn', vmin=-1, vmax=1)
plt.title('Correlation Matrix of Housing Data')
plt.show()
```




```
png

corr_matrix

<div>
```

	median_house_value	median_income	total_rooms	total_be
median_house_value	1.000000	0.691871	0.130991	0.04578
median_income	0.691871	1.000000	0.195383	-0.0134
total_rooms	0.130991	0.195383	1.000000	0.92840
total_bedrooms	0.045783	-0.013495	0.928403	1.00000

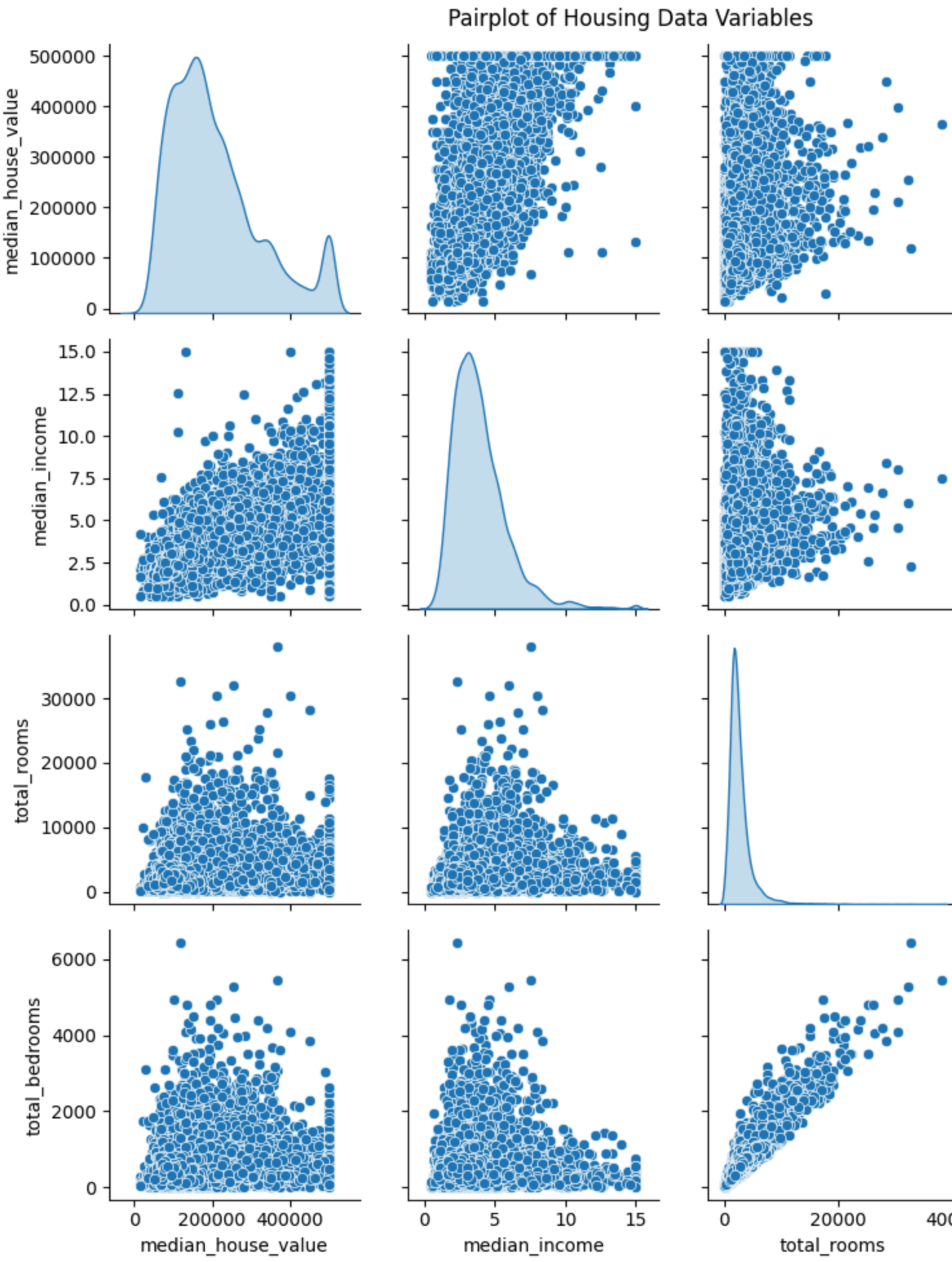
```
<div class="colab-df-buttons">


<button class="colab-df-convert" onclick="convertToInteractive('df-5ad3da5b-a
        title="Convert this dataframe to an interactive table."
        style="display:none;">


<style>
  .colab-df-generate {
    background-color: #E8F0FE;
    border: none;
    border-radius: 50%;
    cursor: pointer;
    display: none;
    fill: #1967D2;
    height: 32px;
    padding: 0 0 0 0;
    width: 32px;
  }

  .colab-df-generate:hover {
    background-color: #E2EBFA;
    box-shadow: 0px 1px 2px rgba(60, 64, 67, 0.3), 0px 1px 3px 1px rgba(60, 6
    fill: #174EA6;
  }

  [theme=dark] .colab-df-generate {
```

```
    background-color: #3B4455;
    fill: #D2E3FC;
  }

  [theme=dark] .colab-df-generate:hover {
    background-color: #434B5C;
    box-shadow: 0px 1px 3px 1px rgba(0, 0, 0, 0.15);
    filter: drop-shadow(0px 1px 2px rgba(0, 0, 0, 0.3));
    fill: #FFFFFF;
  }
</style>
<button class="colab-df-generate" onclick="generateWithVariable('corr_matrix'
        title="Generate code using this dataframe."
        style="display:none;">

</div>
```

The correlation matrix shows the relationships between AQI values of different pollutants, including CO, Ozone, NO2, and PM2.5. It helps us understand the degree of association between these pollutants and how they contribute to the overall air quality index. The correlation values indicate whether the AQI values of different pollutants tend to increase or decrease together.

Scatterplot Matrix of Pollutant AQI Values

```
sns.pairplot(data=california_data, vars=['median_house_value', 'median_income
plt.suptitle('Pairplot of Housing Data Variables')
plt.tight_layout()
plt.show()
```


Pairplot of Housing Data Variables
median_house_value
median_income
total_rooms
total_bedrooms
500000
400000
300000
200000
100000
0
15.0
12.5
10.0
7.5
5.0
2.5
0.0
30000
20000
10000
6000
4000
2000
0
200000
400000
5
10
15
20000
400

png

```
data.columns
```

Top polluted Cities

5. Conclusion

Based on the exploratory data analysis (EDA) conducted on the provided air quality dataset, the following conclusions can be drawn:

- **Overall Air Quality:** The majority of the recorded AQI values fall within the “Good” to “Moderate” categories, indicating relatively acceptable air quality in many locations. However, there are also instances of higher AQI values, indicating poorer air quality.

- **Pollutant Contribution:** Among the pollutants measured, PM2.5, NO2, CO, and Ozone are the major contributors to air pollution. These pollutants show varying levels of correlation with each other, suggesting potential interactions and shared sources.

- **Top Polluted Countries:** The analysis highlights the countries with consistently poor air quality based on the recorded AQI values. Identifying these countries can aid in directing resources and interventions to improve air quality in those regions.

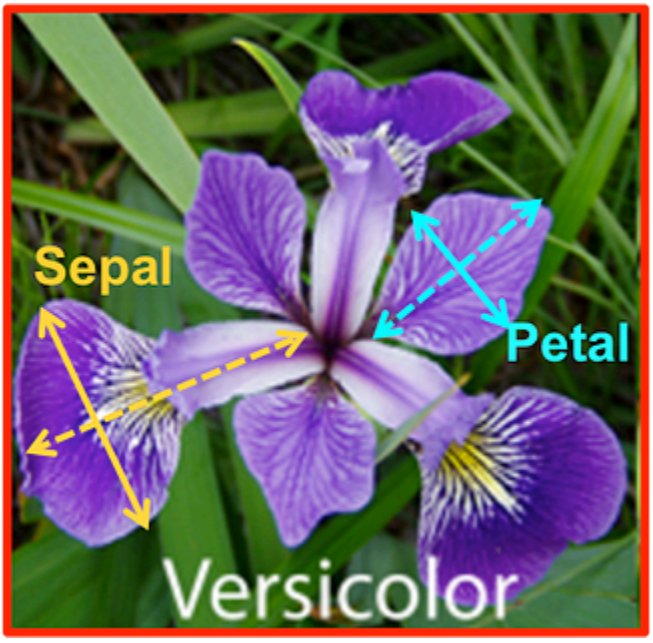
- **Geospatial Analysis:** Visualizing the geographic distribution of air quality using latitude and longitude coordinates helps identify areas with higher pollution levels. Certain regions may exhibit clusters of polluted areas, indicating localized sources of pollution.

In conclusion, the EDA provides valuable insights into air quality, pollutant contributions, geographical patterns, and potential avenues for further analysis. These findings can inform decision-making processes, policy formulation, and targeted interventions aimed at improving air quality and mitiga

```
# IMPORTANT: RUN THIS CELL IN ORDER TO IMPORT YOUR KAGGLE DATA SOURCES,
# THEN FEEL FREE TO DELETE THIS CELL.
# NOTE: THIS NOTEBOOK ENVIRONMENT DIFFERS FROM KAGGLE'S PYTHON
# ENVIRONMENT SO THERE MAY BE MISSING LIBRARIES USED BY YOUR
# NOTEBOOK.
import kagglehub
uciml_iris_path = kagglehub.dataset_download('uciml/iris')

print('Data source import complete.')
```

```
Using Colab cache for faster access to the 'iris' dataset.
```

```
Data source import complete.

# This Python 3 environment comes with many helpful analytics libraries instal
# It is defined by the kaggle/python Docker image: https://github.com/kaggle/d
# For example, here's several helpful packages to load

import numpy as np # linear algebra
import pandas as pd # data processing, CSV file I/O (e.g. pd.read_csv)
import warnings
warnings.filterwarnings('ignore')
# Input data files are available in the read-only "../input/" directory
# For example, running this (by clicking run or pressing Shift+Enter) will lis

import os
for dirname, _, filenames in os.walk('/kaggle/input'):
    for filename in filenames:
        print(os.path.join(dirname, filename))

# You can write up to 20GB to the current directory (/kaggle/working/) that ge
# You can also write temporary files to /kaggle/temp/, but they won't be saved

/kaggle/input/iris/Iris.csv
/kaggle/input/iris/database.sqlite
```



REFERENCES - Some of the graph ideas in this kernel are taken from https://avinashkz.com/data/visualization/2017/09/28/seaborn.html

Dataset overview

The data set consists of 50 samples from each of three species Iris setosa, Iris virginica and Iris versicolor. 4 features were measured from each sample: the length and the width of the sepals and petals, in centimeters.

```
df = pd.read_csv('/kaggle/input/iris/Iris.csv')
df.head()
```

	Id	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm	Species
0	1	5.1	3.5	1.4	0.2	Iris-setosa
1	2	4.9	3.0	1.4	0.2	Iris-setosa
2	3	4.7	3.2	1.3	0.2	Iris-setosa
3	4	4.6	3.1	1.5	0.2	Iris-setosa
4	5	5.0	3.6	1.4	0.2	Iris-setosa

```
<div class="colab-df-buttons">


<button class="colab-df-convert" onclick="convertToInteractive('df-d47e5559-4
        title="Convert this dataframe to an interactive table."
        style="display:none;">
```

```
</div>

#make the Id column as index
df = pd.read_csv('/kaggle/input/iris/Iris.csv', index_col='Id')
df.tail()
```

	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm	Species
Id					
146	6.7	3.0	5.2	2.3	Iris-virginica
147	6.3	2.5	5.0	1.9	Iris-virginica
148	6.5	3.0	5.2	2.0	Iris-virginica
149	6.2	3.4	5.4	2.3	Iris-virginica
150	5.9	3.0	5.1	1.8	Iris-virginica

```
<div class="colab-df-buttons">


<button class="colab-df-convert" onclick="convertToInteractive('df-8050fd9a-2
        title="Convert this dataframe to an interactive table."
        style="display:none;">
```

```
</div>

#Rename columns
df.rename({'SepalLengthCm': 'sepal_length',
           'SepalWidthCm': 'sepal_width',
           'PetalLengthCm': 'petal_length',
           'PetalWidthCm': 'petal_width',
           'Species': 'species'
          }, inplace=True,axis=1)
df.sample(5)
```

	sepal_length	sepal_width	petal_length	petal_width	species
Id					
122	5.6	2.8	4.9	2.0	Iris-virginica
109	6.7	2.5	5.8	1.8	Iris-virginica
48	4.6	3.2	1.4	0.2	Iris-setosa
5	5.0	3.6	1.4	0.2	Iris-setosa
10	4.9	3.1	1.5	0.1	Iris-setosa

```
<div class="colab-df-buttons">


<button class="colab-df-convert" onclick="convertToInteractive('df-e3ebe57c-5
        title="Convert this dataframe to an interactive table."
        style="display:none;">
```

```
</div>

#calculate median of each species
x = df.groupby('species').median()
x
```

	sepal_length	sepal_width	petal_length	petal_width
species				
Iris-setosa	5.0	3.4	1.50	0.2
Iris-versicolor	5.9	2.8	4.35	1.3
Iris-virginica	6.5	3.0	5.55	2.0

```
<div class="colab-df-buttons">


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        style="display:none;">
```

```
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    border: none;
    border-radius: 50%;
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    display: none;
    fill: #1967D2;
    height: 32px;
    padding: 0 0 0 0;
    width: 32px;
  }

  .colab-df-generate:hover {
    background-color: #E2EBFA;
    box-shadow: 0px 1px 2px rgba(60, 64, 67, 0.3), 0px 1px 3px 1px rgba(60, 64
    fill: #174EA6;
  }

  [theme=dark] .colab-df-generate {
    background-color: #3B4455;
    fill: #D2E3FC;
  }

  [theme=dark] .colab-df-generate:hover {
    background-color: #434B5C;
    box-shadow: 0px 1px 3px 1px rgba(0, 0, 0, 0.15);
    filter: drop-shadow(0px 1px 2px rgba(0, 0, 0, 0.3));
    fill: #FFFFFF;
  }
</style>
<button class="colab-df-generate" onclick="generateWithVariable('x')"
        title="Generate code using this dataframe."
        style="display:none;">
```

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```
</div>

#Statistics of this dataset
stats = df.describe()
stats
```

	sepal_length	sepal_width	petal_length	petal_width
count	150.000000	150.000000	150.000000	150.000000
mean	5.843333	3.054000	3.758667	1.198667
std	0.828066	0.433594	1.764420	0.763161
min	4.300000	2.000000	1.000000	0.100000
25%	5.100000	2.800000	1.600000	0.300000
50%	5.800000	3.000000	4.350000	1.300000
75%	6.400000	3.300000	5.100000	1.800000
max	7.900000	4.400000	6.900000	2.500000

```
<div class="colab-df-buttons">


<button class="colab-df-convert" onclick="convertToInteractive('df-b1c8fab7-6
        title="Convert this dataframe to an interactive table."
        style="display:none;">
```

```
<style>
  .colab-df-generate {
```

```
    background-color: #E8F0FE;
    border: none;
    border-radius: 50%;
    cursor: pointer;
    display: none;
    fill: #1967D2;
    height: 32px;
    padding: 0 0 0 0;
    width: 32px;
  }

  .colab-df-generate:hover {
    background-color: #E2EBFA;
    box-shadow: 0px 1px 2px rgba(60, 64, 67, 0.3), 0px 1px 3px 1px rgba(60, 64
    fill: #174EA6;
  }

  [theme=dark] .colab-df-generate {
    background-color: #3B4455;
    fill: #D2E3FC;
  }

  [theme=dark] .colab-df-generate:hover {
    background-color: #434B5C;
    box-shadow: 0px 1px 3px 1px rgba(0, 0, 0, 0.15);
    filter: drop-shadow(0px 1px 2px rgba(0, 0, 0, 0.3));
    fill: #FFFFFF;
  }
</style>
<button class="colab-df-generate" onclick="generateWithVariable('stats')"
        title="Generate code using this dataframe."
        style="display:none;">

🪄

</div>

#calculate range, and move columns 25%,50%,75%  and mean values into a datafra
stats.loc['range'] = stats.loc['max'] - stats.loc['min']
new_df = stats.loc[['mean', 'range', '25%', '50%', '75%']]
new_df
```

	sepal_length	sepal_width	petal_length	petal_width
mean	5.843333	3.054	3.758667	1.198667
range	3.600000	2.400	5.900000	2.400000
25%	5.100000	2.800	1.600000	0.300000

	sepal_length	sepal_width	petal_length	petal_width
median	5.800000	3.000	4.350000	1.300000
75%	6.400000	3.300	5.100000	1.800000

```
<div class="colab-df-buttons">


<button class="colab-df-convert" onclick="convertToInteractive('df-3927c108-6a
        title="Convert this dataframe to an interactive table."
        style="display:none;">
```

```
<style>
  .colab-df-generate {
    background-color: #E8F0FE;
    border: none;
    border-radius: 50%;
    cursor: pointer;
    display: none;
    fill: #1967D2;
    height: 32px;
    padding: 0 0 0 0;
    width: 32px;
  }

  .colab-df-generate:hover {
    background-color: #E2EBFA;
```

```
    box-shadow: 0px 1px 2px rgba(60, 64, 67, 0.3), 0px 1px 3px 1px rgba(60, 64
    fill: #174EA6;
  }

  [theme=dark] .colab-df-generate {
    background-color: #3B4455;
    fill: #D2E3FC;
  }

  [theme=dark] .colab-df-generate:hover {
    background-color: #434B5C;
    box-shadow: 0px 1px 3px 1px rgba(0, 0, 0, 0.15);
    filter: drop-shadow(0px 1px 2px rgba(0, 0, 0, 0.3));
    fill: #FFFFFF;
  }
</style>
<button class="colab-df-generate" onclick="generateWithVariable('new_df')"
        title="Generate code using this dataframe."
        style="display:none;">
```

```
</div>
```

Aggregation functions

```
#Calculate mean and median or others using multiple functions at once - 2 meth

df.groupby('species').agg(['mean', 'median'])  # passing a list of recognized
df.groupby('species').agg([np.mean, np.median])
```

```
<div>
```

	sepal_length		sepal_width		petal_length		petal_width	
	mean	median	mean	median	mean	median	mean	median
species								
Iris-setosa	5.006	5.0	3.418	3.4	1.464	1.50	0.244	0.2
Iris-versicolor	5.936	5.9	2.770	2.8	4.260	4.35	1.326	1.3
Iris-virginica	6.588	6.5	2.974	3.0	5.552	5.55	2.026	2.0

```
<div class="colab-df-buttons">


<button class="colab-df-convert" onclick="convertToInteractive('df-dc2b6fba-a
        title="Convert this dataframe to an interactive table."
```

```
        style="display:none;">
```

```
</div>

#If we want an explicit aggr function like max for a column
agg_dict={field: ['mean', 'median'] for  field in df.columns.tolist() if fiel
agg_dict['petal_length'] = 'max'
df.groupby('species').agg(agg_dict)
```

	sepal_length		sepal_width		petal_length	petal_width	
	mean	median	mean	median	max	mean	median
species							
Iris-setosa	5.006	5.0	3.418	3.4	1.9	0.244	0.2
Iris-versicolor	5.936	5.9	2.770	2.8	5.1	1.326	1.3
Iris-virginica	6.588	6.5	2.974	3.0	6.9	2.026	2.0

```
<div class="colab-df-buttons">


<button class="colab-df-convert" onclick="convertToInteractive('df-d59810e6-5
        title="Convert this dataframe to an interactive table."
        style="display:none;">
```

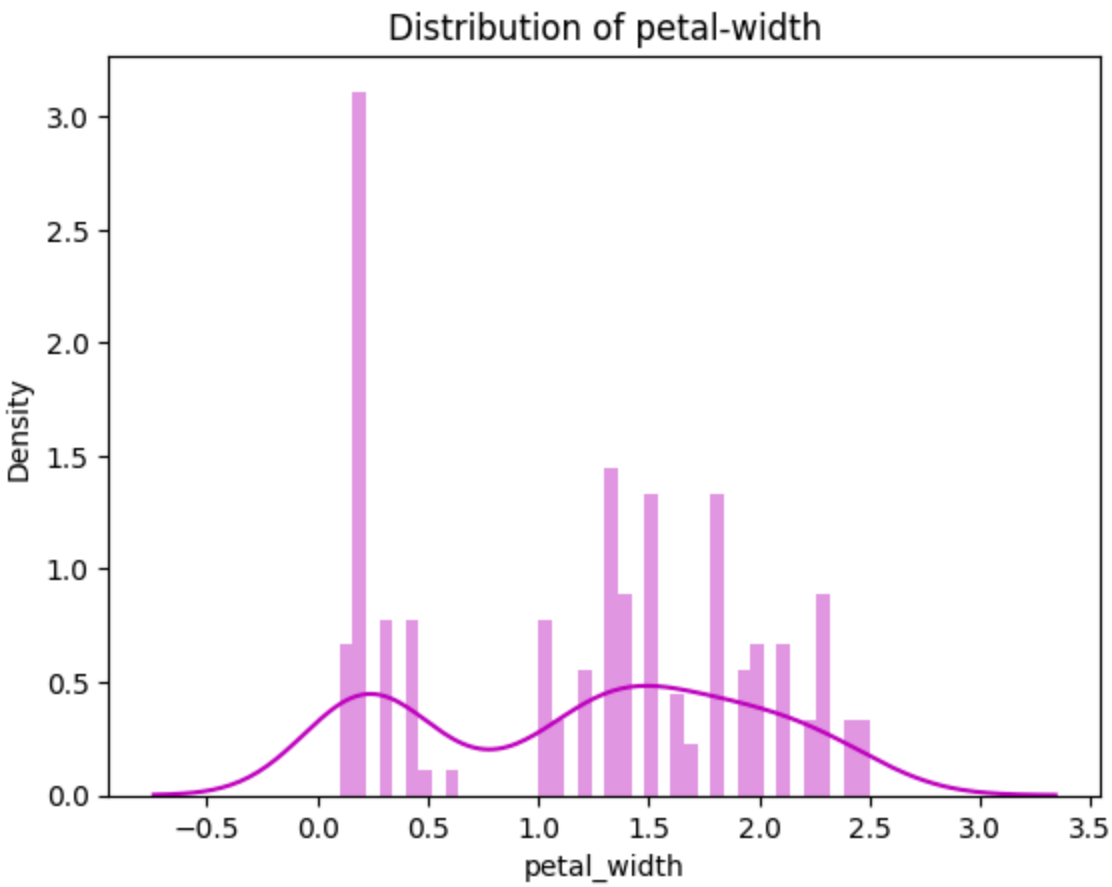
Visualisations

```
import matplotlib.pyplot as plt
%matplotlib inline
import seaborn as sns
```

Distribution Plot

```
sns.distplot(a=df['petal_width'], bins=40, color='m')
plt.title('Distribution of petal-width')
```

```
Text(0.5, 1.0, 'Distribution of petal-width')
```



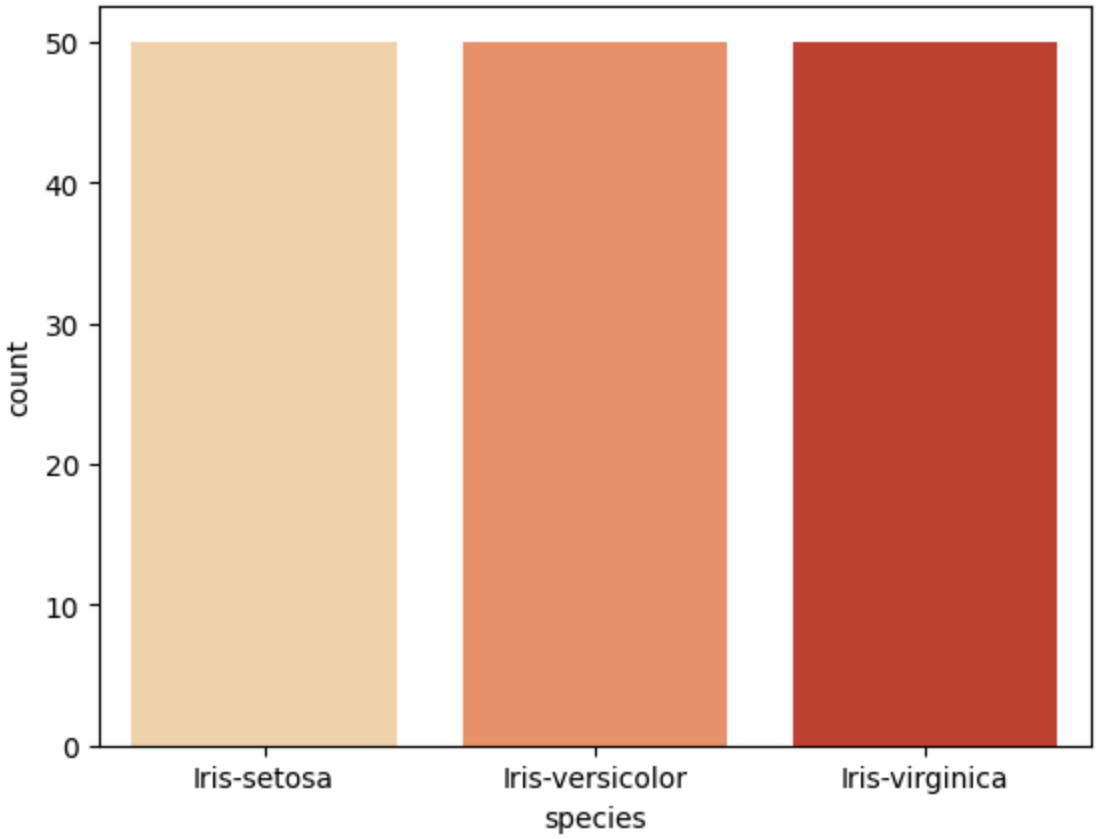
png

Count plot

displays the number of observations for a categorical variable using bars.

```
sns.countplot(x='species',data=df, palette="OrRd")
```

```
<Axes: xlabel='species', ylabel='count'>
```



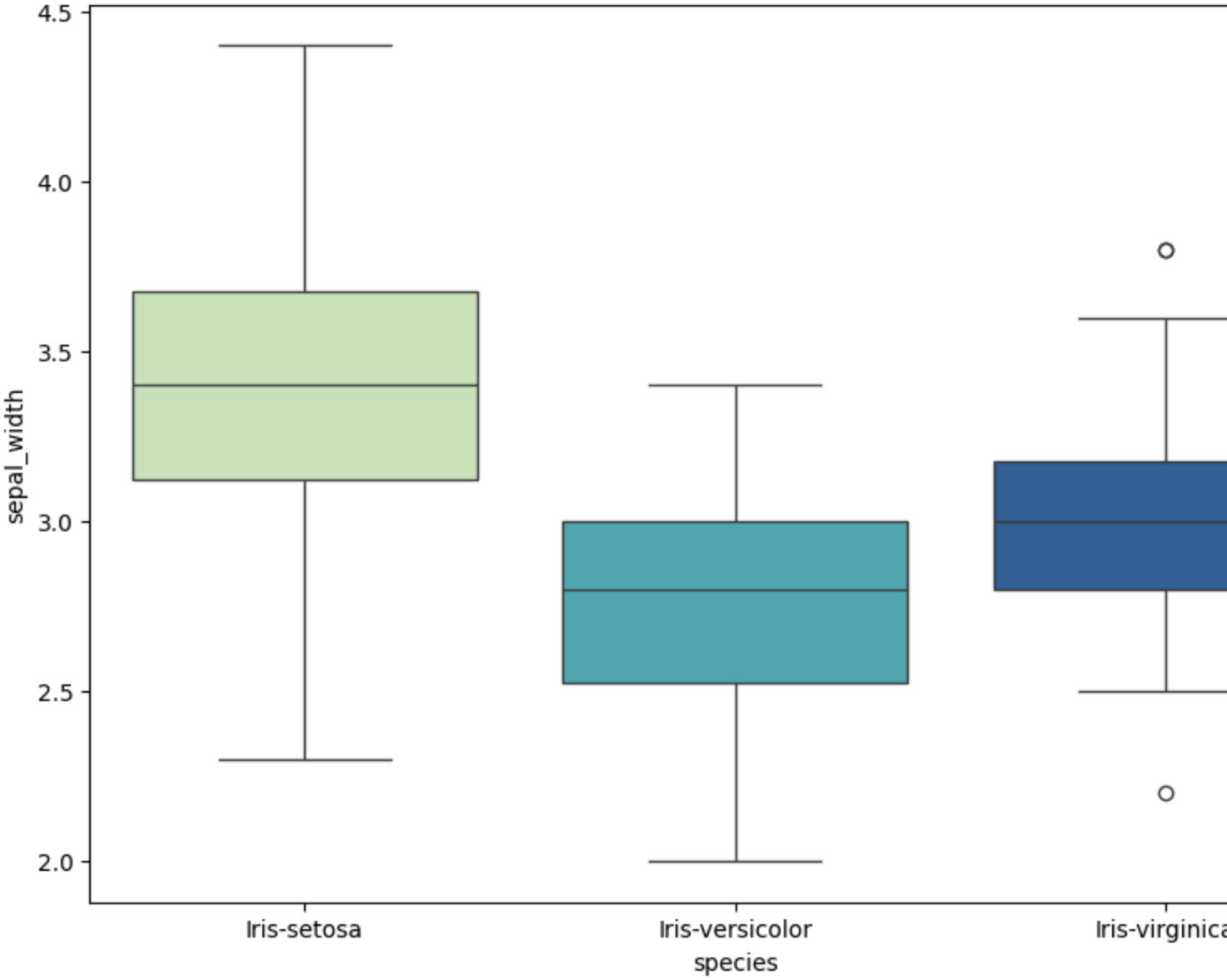
png

Box Plot

shows the distribution of quantitative data across a categorical variable.

```
plt.figure(figsize=(10,7))
sns.boxplot(x='species',y='sepal_width',data=df ,palette='YlGnBu')
```

```
<Axes: xlabel='species', ylabel='sepal_width'>
```



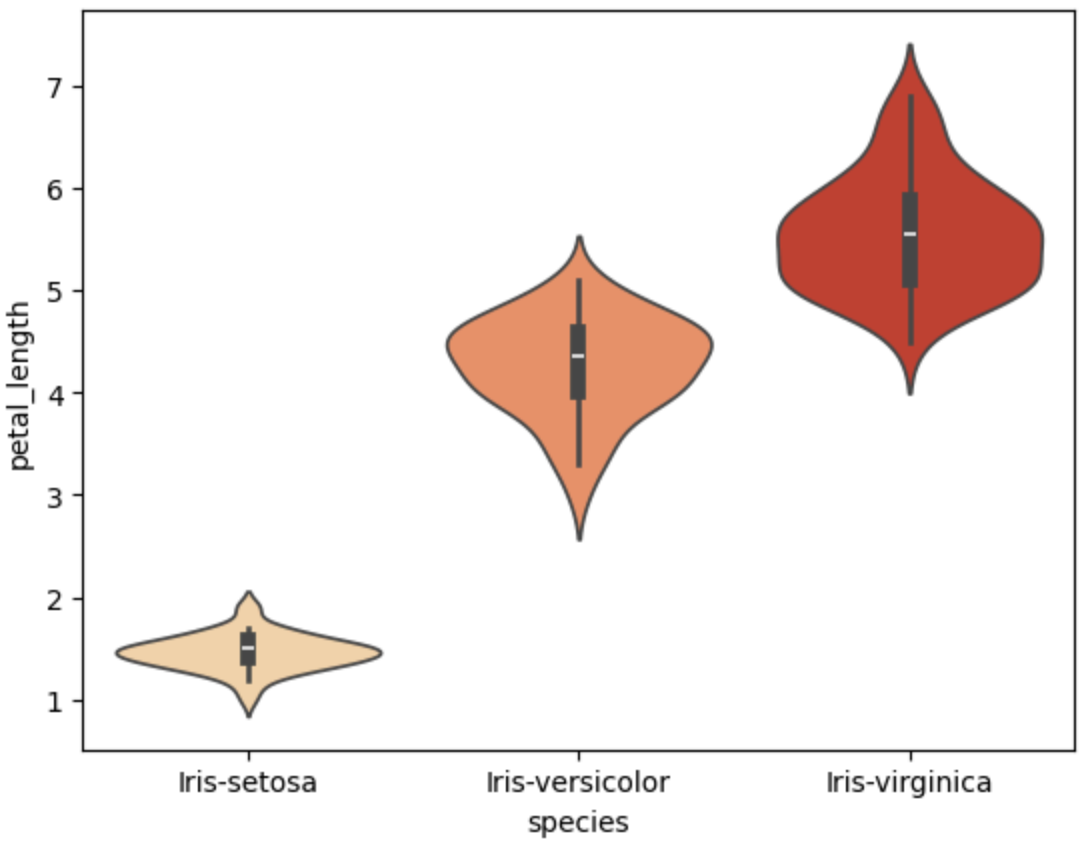
png

violin plot

This is a variation of box plot. It shows the distribution of quantitative data across several levels of categorical variables. The violin plot also features a kde of the underlying distribution

```
sns.violinplot(x='species', y='petal_length', data=df, palette='OrRd')
```

```
<Axes: xlabel='species', ylabel='petal_length'>
```



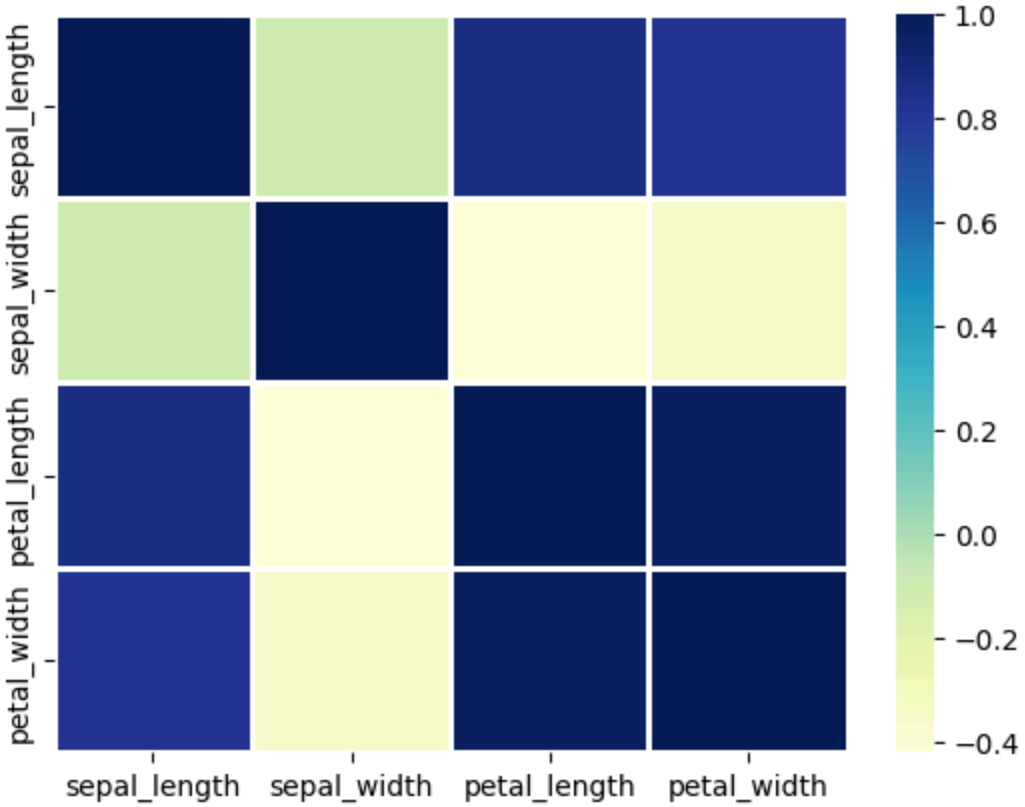
png

Matrix plots

1. HeatMap - It provides the graph of the correlation matrix.
2. Cluster Map - It is a plot of matrix dataset as a hierarchically clustered heatmap.

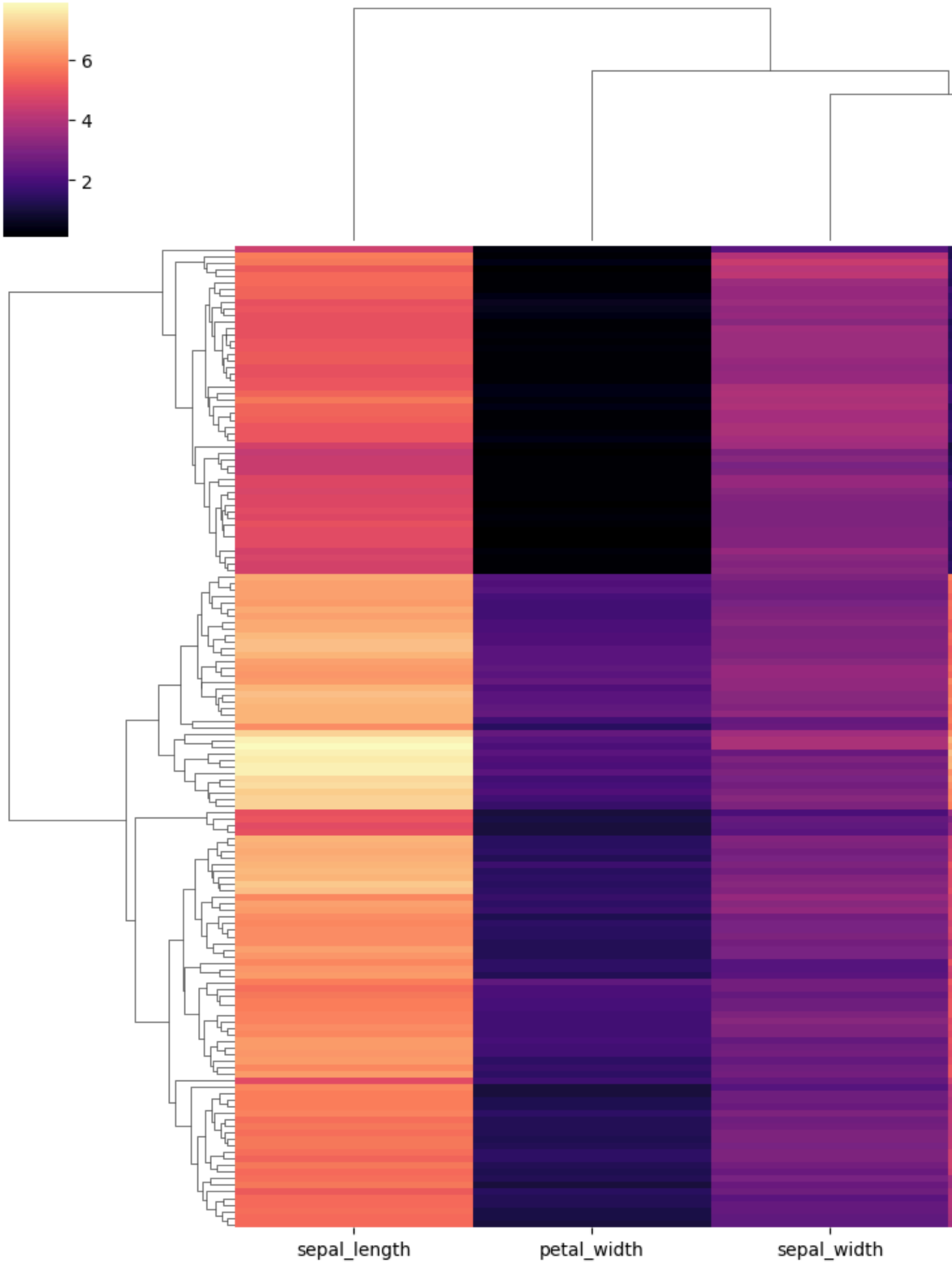
```
sns.heatmap(df.drop('species', axis=1).corr(),cmap="YlGnBu", linecolor='white
```

```
<Axes: >
```



png

```
#Cluster map takes only quantitative variables, so we need to do some processi
x = df.drop('species',axis=1)
g = sns.clustermap(x, cmap="magma")
```

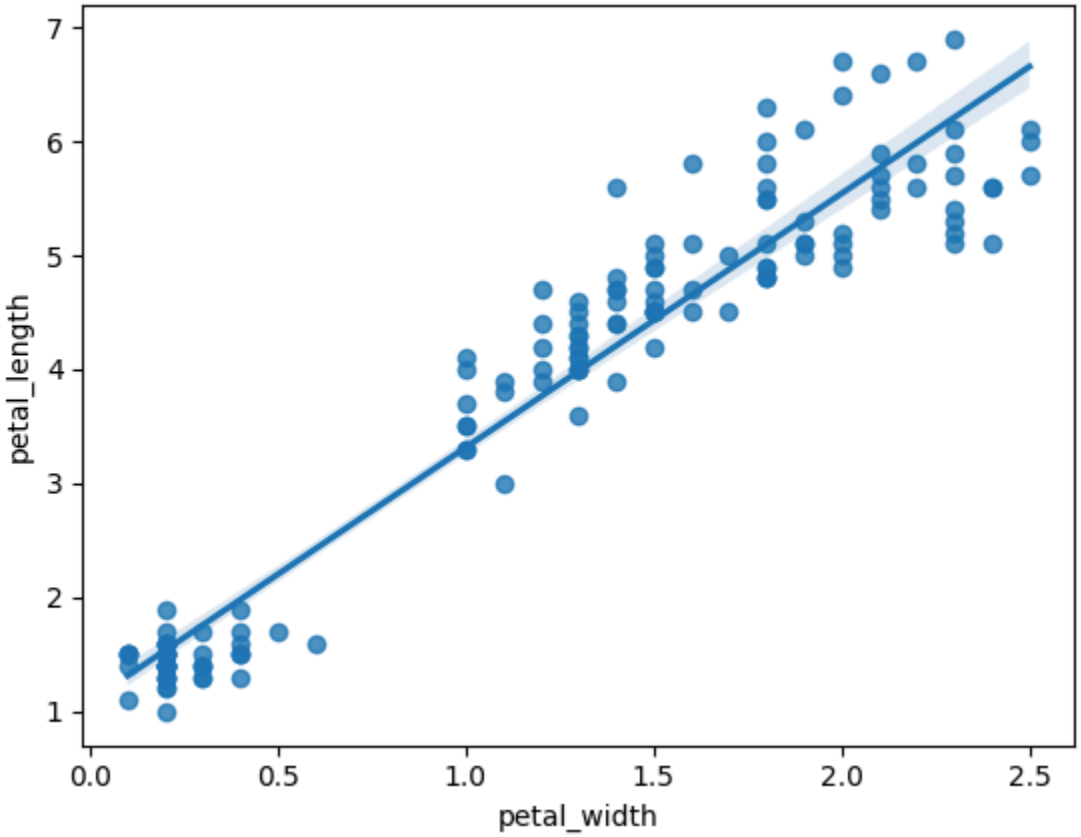

6
4
2
sepal_length
petal_width
sepal_width

png

Regression Plots

```
sns.regplot(x='petal_width', y='petal_length', data=df)
```

```
<Axes: xlabel='petal_width', ylabel='petal_length'>
```



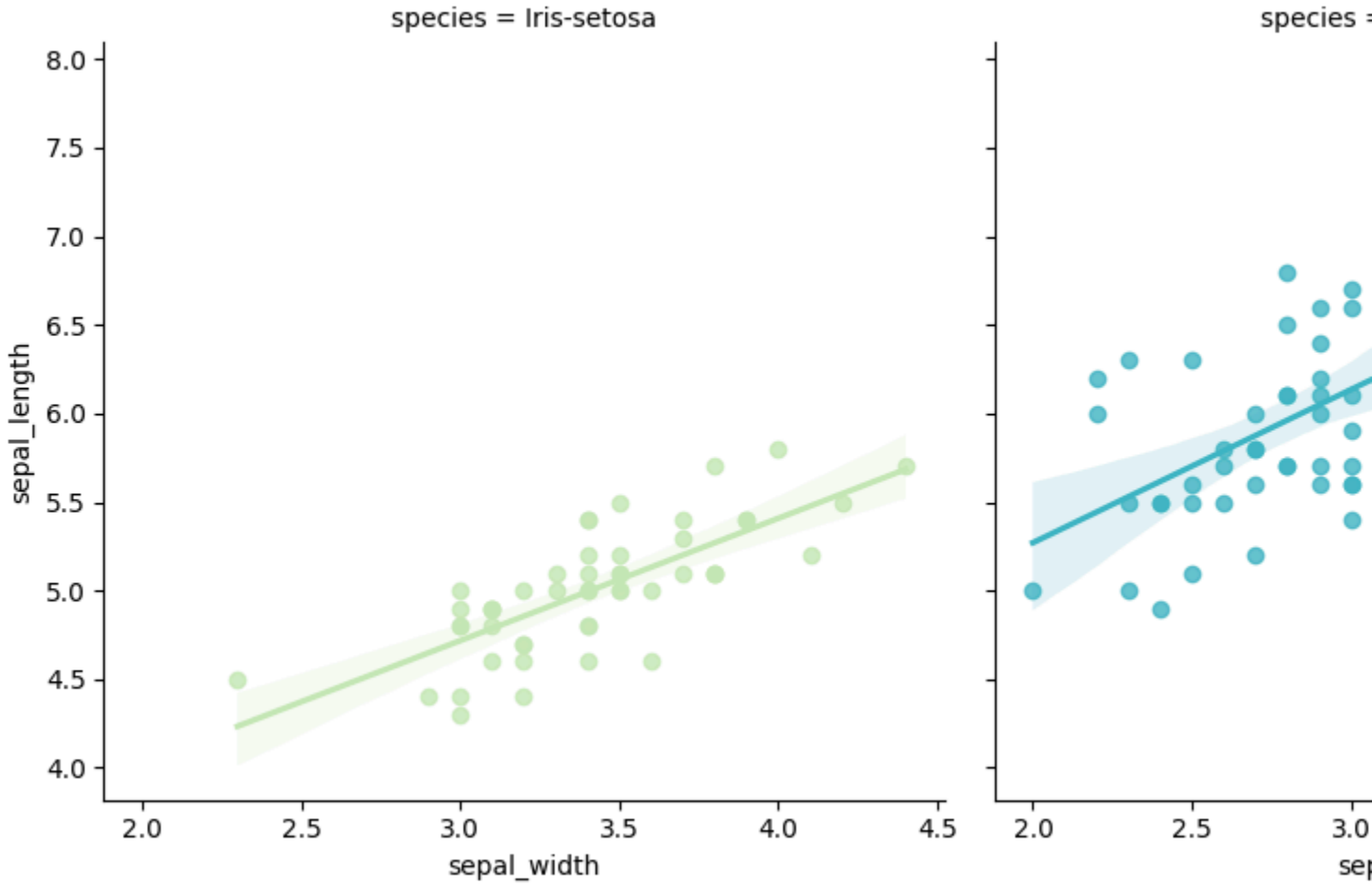
png

Axis Grids - These are graphs that combine various plots using FacetGrid to realize complex visualizations.

LM Plots

```
sns.lmplot(x = 'sepal_width', y = 'sepal_length', data = df, col = 'species',
```

```
<seaborn.axisgrid.FacetGrid at 0x7e0bbceff320>
```



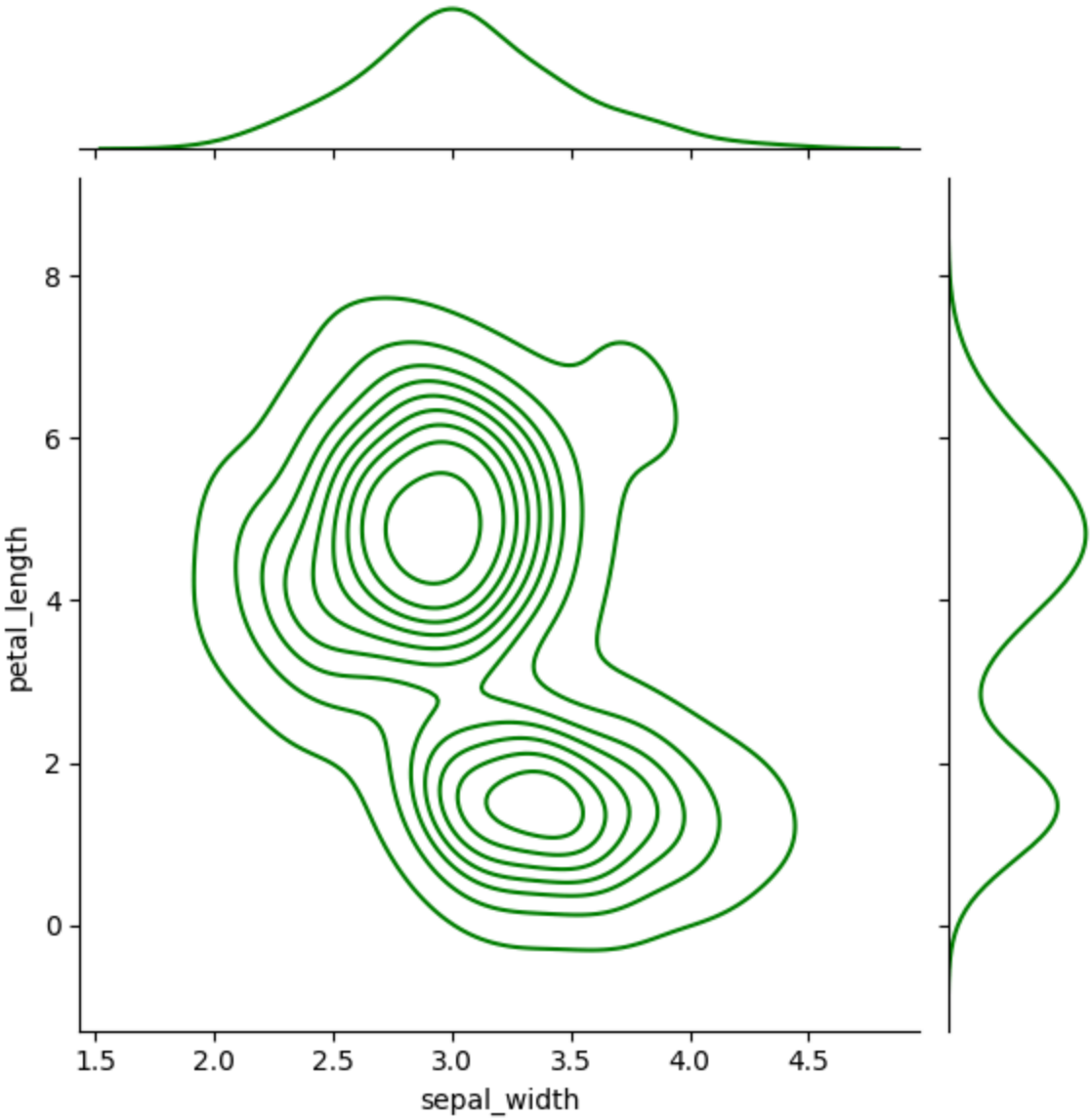
png

Joint Plot

Used to draw a plot of two variables with bivariate and univariate graphs.

```
sns.jointplot(x="sepal_width", y="petal_length", data=df, kind="kde", color="
```

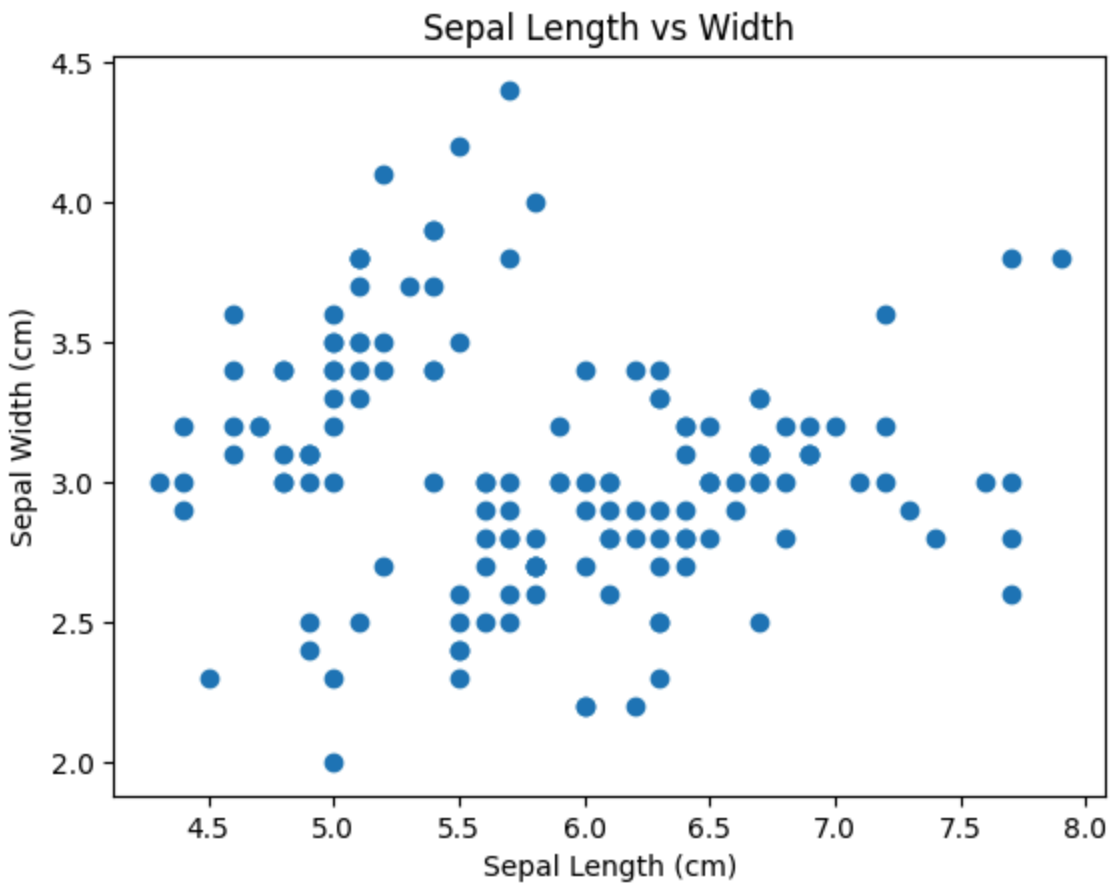
```
<seaborn.axisgrid.JointGrid at 0x7e0bbcb77f80>
```



png

Scatterplot

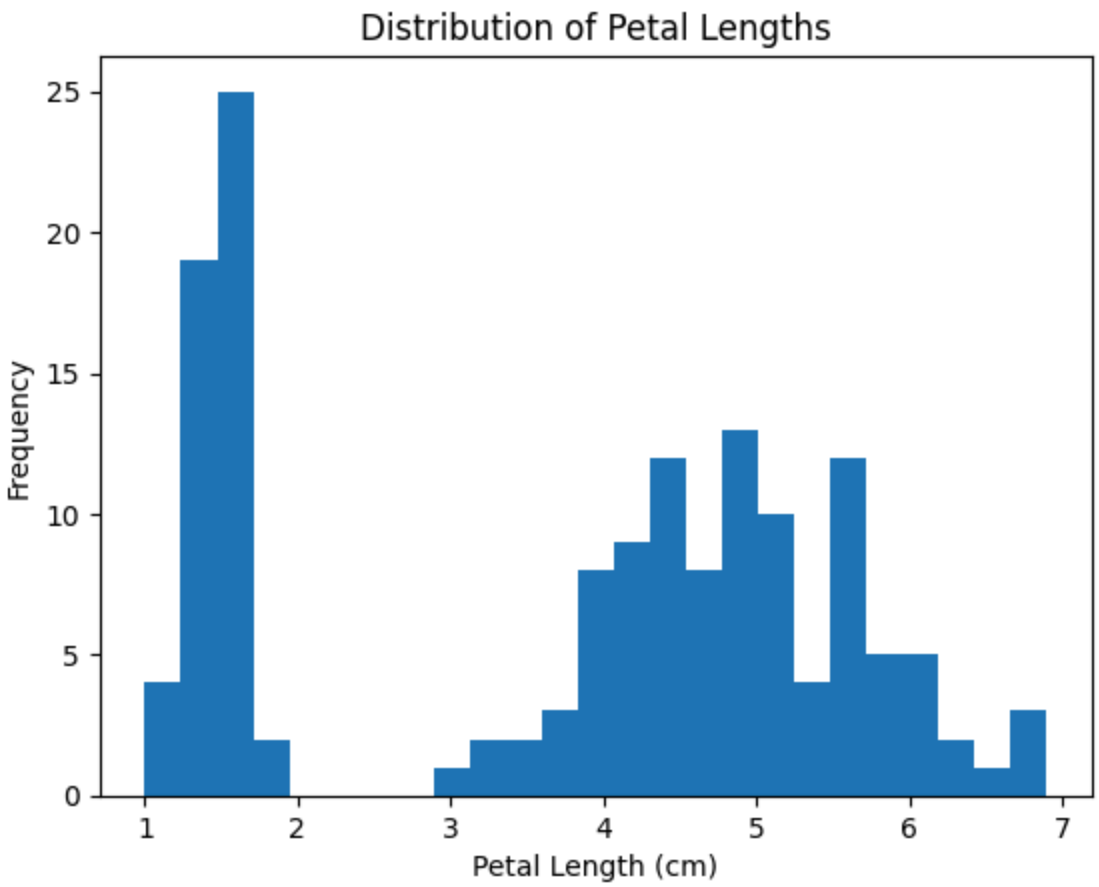
```
ax = plt.axes()

ax.scatter(df.sepal_length, df.sepal_width)

# Label the axes
ax.set(xlabel='Sepal Length (cm)',
       ylabel='Sepal Width (cm)',
       title='Sepal Length vs Width');
```



png

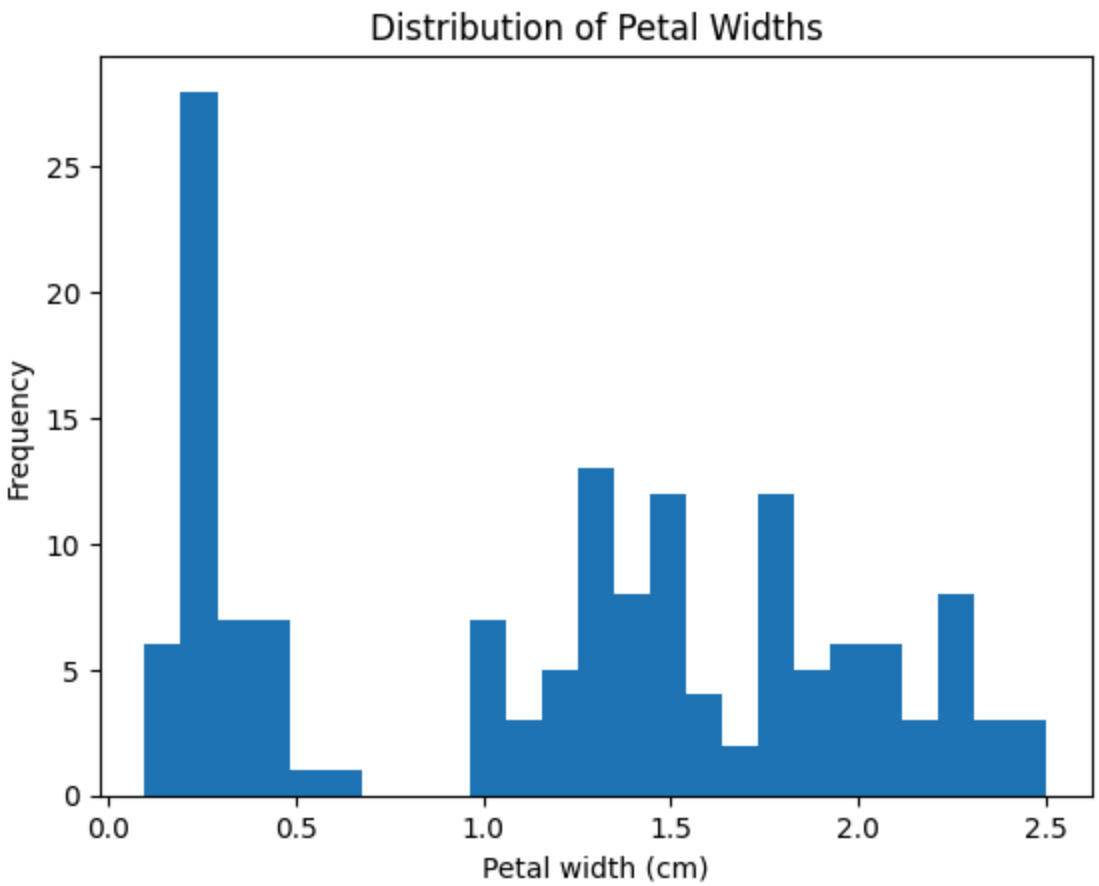
Histograms

```
ax = plt.axes()
ax.hist(df.petal_length, bins=25);

ax.set(xlabel='Petal Length (cm)',
       ylabel='Frequency',
       title='Distribution of Petal Lengths');
```



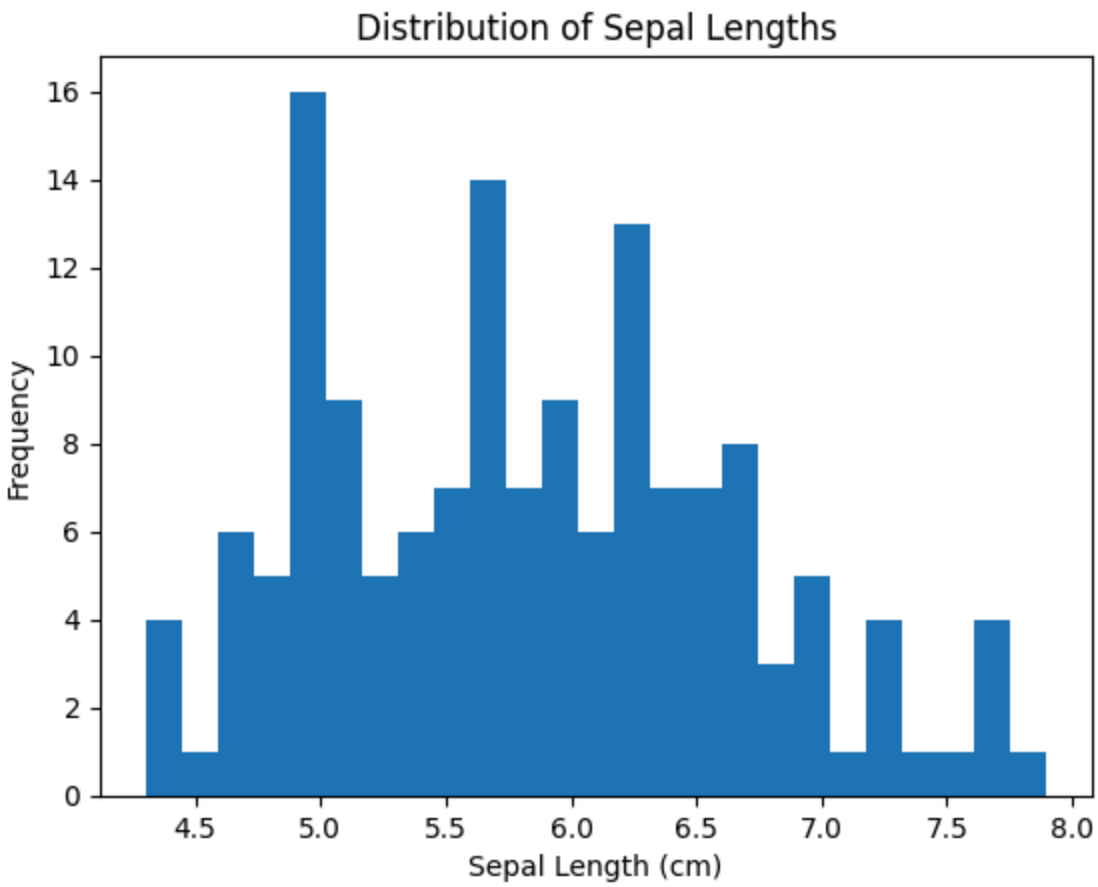
png

```
ax = plt.axes()
ax.hist(df.petal_width, bins=25);

ax.set(xlabel='Petal width (cm)',
       ylabel='Frequency',
       title='Distribution of Petal Widths');
```



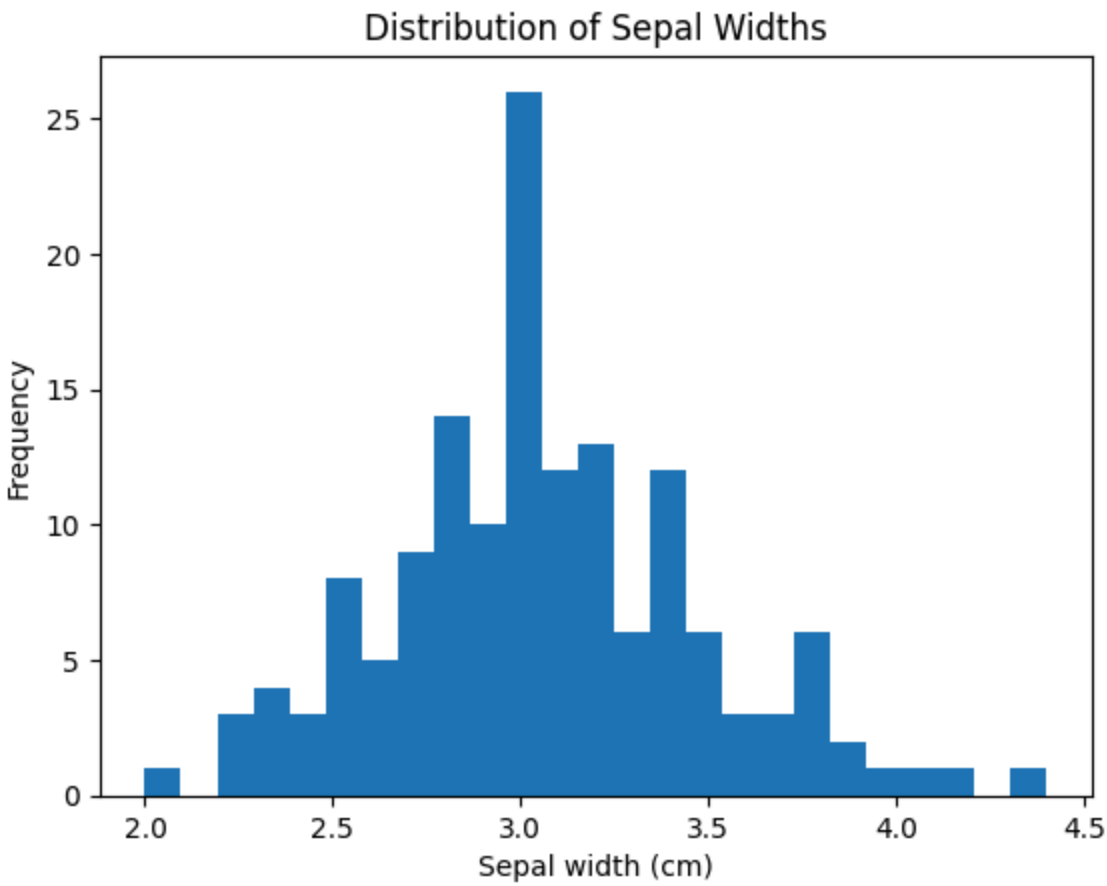
png

```
ax = plt.axes()
ax.hist(df.sepal_length, bins=25);

ax.set(xlabel='Sepal Length (cm)',
       ylabel='Frequency',
       title='Distribution of Sepal Lengths');
```



png

```
ax = plt.axes()
ax.hist(df.sepal_width, bins=25);

ax.set(xlabel='Sepal width (cm)',
       ylabel='Frequency',
       title='Distribution of Sepal Widths');
```

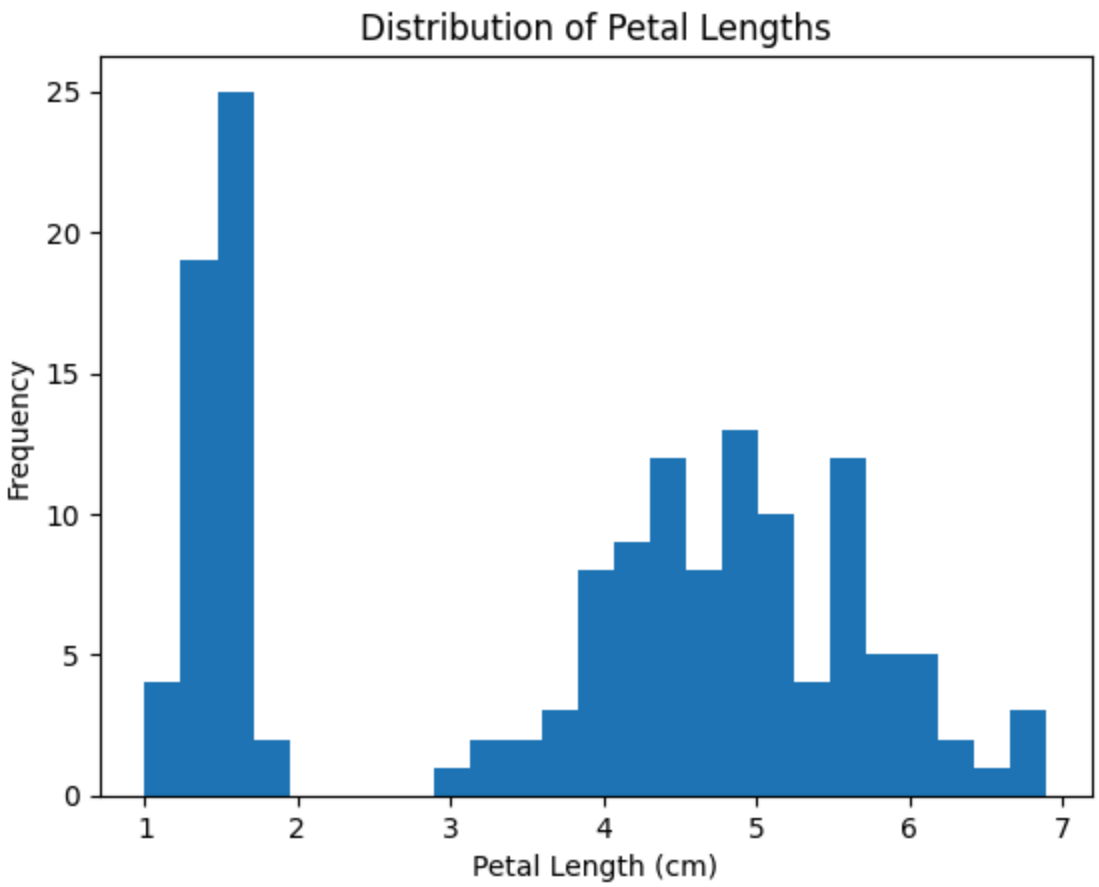


png

TAKE-AWAY From the histograms, we observe that Petal length and petal width are influencing variables for the different species

plot histogram using pandas

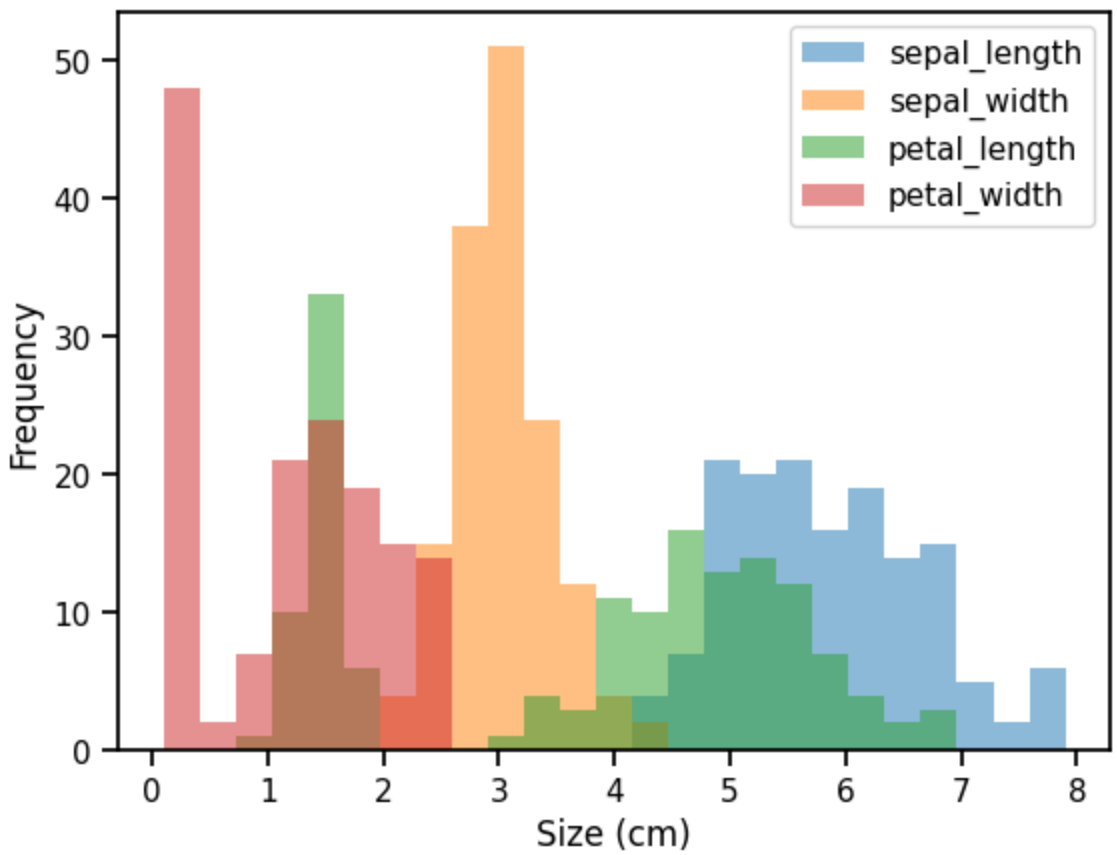
```
ax = df.petal_length.plot.hist(bins=25)

ax.set(xlabel='Petal Length (cm)',
       ylabel='Frequency',
       title='Distribution of Petal Lengths');
```



png

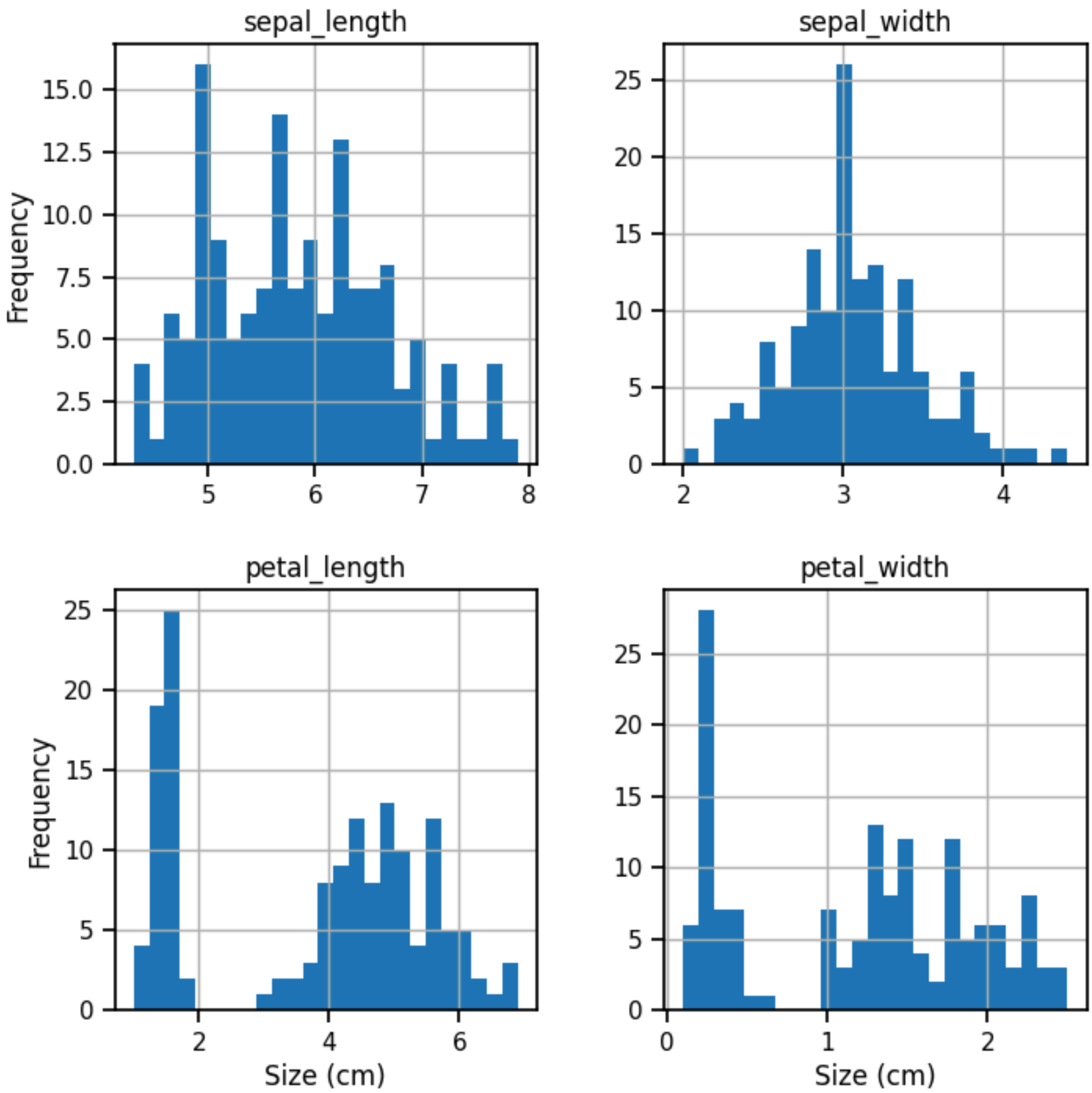
plotting all histograms

```
import seaborn as sns
sns.set_context('notebook')


ax = df.plot.hist(bins=25, alpha=0.5)
ax.set_xlabel('Size (cm)');
```



png

```
axList = df.hist(bins=25, figsize=(8,8))

# Add some x- and y- labels to first column and last row
# Get the number of rows and columns in the subplot grid
num_rows, num_cols = axList.shape

for i in range(num_rows):
    for j in range(num_cols):
        ax = axList[i, j]
        if i == num_rows - 1:  # Check if it's the last row
            ax.set_xlabel('Size (cm)')

        if j == 0:  # Check if it's the first column
            ax.set_ylabel('Frequency')
```



png

all features in single boxplot

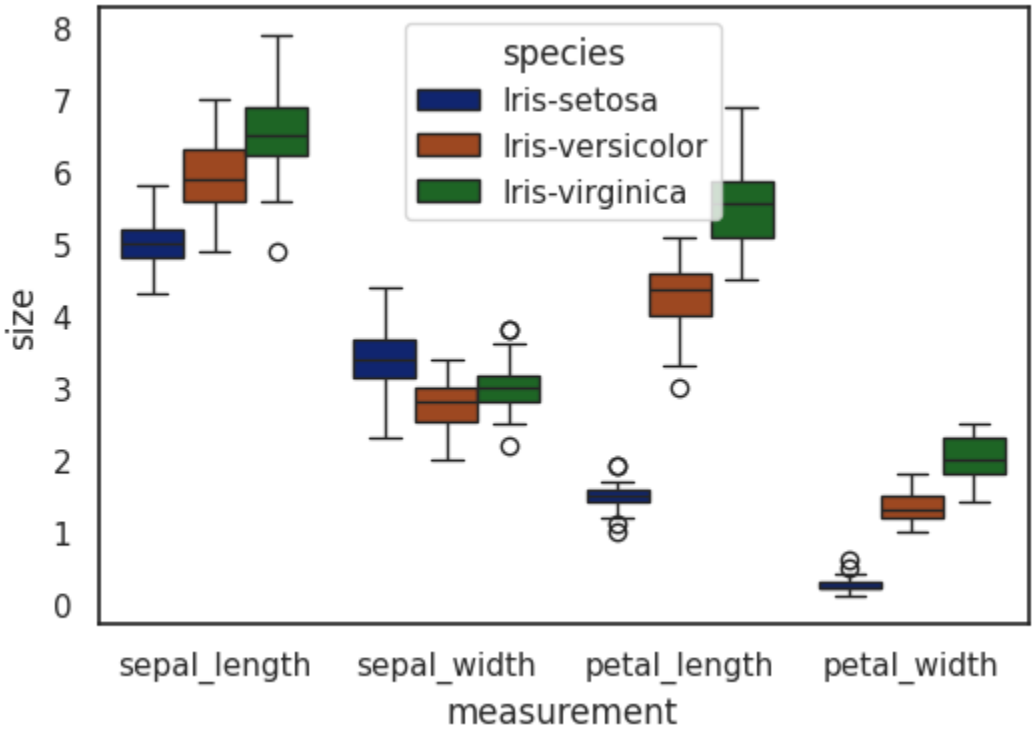
```
# First we have to reshape the data so there is
# only a single measurement in each column

plot_data = (df
             .set_index('species')
```

```
              .stack()
              .to_frame()
              .reset_index()
              .rename(columns={0:'size', 'level_1':'measurement'})
             )

plot_data.head()
```

	species	measurement	size
0	Iris-setosa	sepal_length	5.1
1	Iris-setosa	sepal_width	3.5
2	Iris-setosa	petal_length	1.4
3	Iris-setosa	petal_width	0.2
4	Iris-setosa	sepal_length	4.9

```
<div class="colab-df-buttons">


<button class="colab-df-convert" onclick="convertToInteractive('df-36466e68-b0
        title="Convert this dataframe to an interactive table."
        style="display:none;">
```

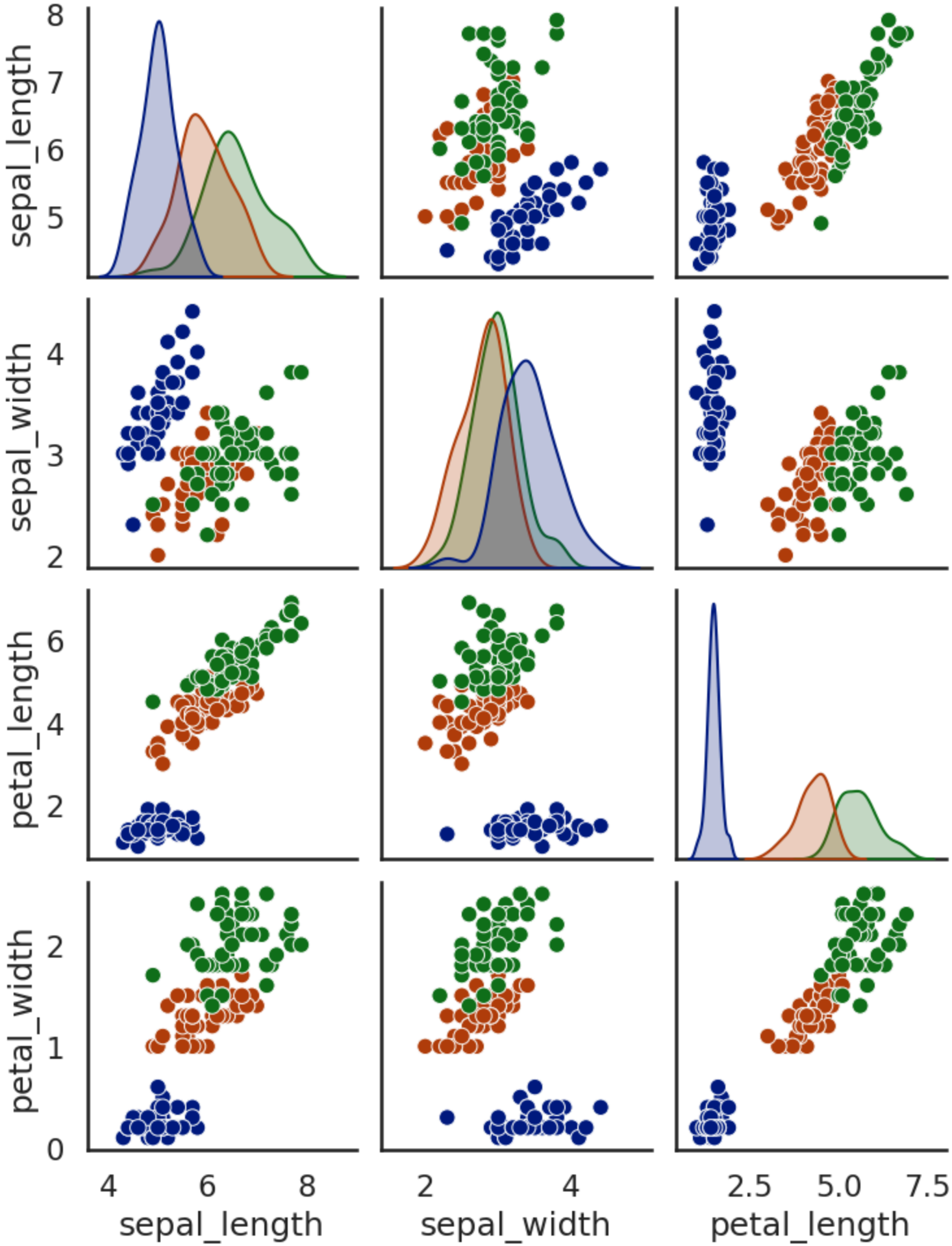
```
</div>

sns.set_style('white')
sns.set_context('notebook')
sns.set_palette('dark')

f = plt.figure(figsize=(6,4))
sns.boxplot(x='measurement', y='size',
            hue='species', data=plot_data);
```



png

Pairplot to examine the correlations

```
sns.set_context('talk')
sns.pairplot(df, hue='species');
```


sepal_length
sepal_width
petal_length
petal_width
8
7
6
5
4
3
2
6
4
2
2
1
0
4
6
8
2
4
2.5
5.0
7.5
sepal_length
sepal_width
petal_length

png

```
from sklearn import datasets
from sklearn.cluster import KMeans
#load the data
iris = datasets.load_iris()

print(iris.data)
```

```
[[5.1 3.5 1.4 0.2]
 [4.9 3.  1.4 0.2]
 [4.7 3.2 1.3 0.2]
 [4.6 3.1 1.5 0.2]
 [5.  3.6 1.4 0.2]
 [5.4 3.9 1.7 0.4]
 [4.6 3.4 1.4 0.3]
 [5.  3.4 1.5 0.2]
 [4.4 2.9 1.4 0.2]
 [4.9 3.1 1.5 0.1]
 [5.4 3.7 1.5 0.2]
 [4.8 3.4 1.6 0.2]
 [4.8 3.  1.4 0.1]
 [4.3 3.  1.1 0.1]
 [5.8 4.  1.2 0.2]
 [5.7 4.4 1.5 0.4]
 [5.4 3.9 1.3 0.4]
 [5.1 3.5 1.4 0.3]
 [5.7 3.8 1.7 0.3]
 [5.1 3.8 1.5 0.3]
 [5.4 3.4 1.7 0.2]
 [5.1 3.7 1.5 0.4]
 [4.6 3.6 1.  0.2]
 [5.1 3.3 1.7 0.5]
 [4.8 3.4 1.9 0.2]
 [5.  3.  1.6 0.2]
 [5.  3.4 1.6 0.4]
 [5.2 3.5 1.5 0.2]
 [5.2 3.4 1.4 0.2]
 [4.7 3.2 1.6 0.2]
 [4.8 3.1 1.6 0.2]
 [5.4 3.4 1.5 0.4]
 [5.2 4.1 1.5 0.1]
 [5.5 4.2 1.4 0.2]
 [4.9 3.1 1.5 0.2]
 [5.  3.2 1.2 0.2]
 [5.5 3.5 1.3 0.2]
 [4.9 3.6 1.4 0.1]
 [4.4 3.  1.3 0.2]
 [5.1 3.4 1.5 0.2]
```

```
[5.  3.5 1.3 0.3]
[4.5 2.3 1.3 0.3]
[4.4 3.2 1.3 0.2]
[5.  3.5 1.6 0.6]
[5.1 3.8 1.9 0.4]
[4.8 3.  1.4 0.3]
[5.1 3.8 1.6 0.2]
[4.6 3.2 1.4 0.2]
[5.3 3.7 1.5 0.2]
[5.  3.3 1.4 0.2]
[7.  3.2 4.7 1.4]
[6.4 3.2 4.5 1.5]
[6.9 3.1 4.9 1.5]
[5.5 2.3 4.  1.3]
[6.5 2.8 4.6 1.5]
[5.7 2.8 4.5 1.3]
[6.3 3.3 4.7 1.6]
[4.9 2.4 3.3 1. ]
[6.6 2.9 4.6 1.3]
[5.2 2.7 3.9 1.4]
[5.  2.  3.5 1. ]
[5.9 3.  4.2 1.5]
[6.  2.2 4.  1. ]
[6.1 2.9 4.7 1.4]
[5.6 2.9 3.6 1.3]
[6.7 3.1 4.4 1.4]
[5.6 3.  4.5 1.5]
[5.8 2.7 4.1 1. ]
[6.2 2.2 4.5 1.5]
[5.6 2.5 3.9 1.1]
[5.9 3.2 4.8 1.8]
[6.1 2.8 4.  1.3]
[6.3 2.5 4.9 1.5]
[6.1 2.8 4.7 1.2]
[6.4 2.9 4.3 1.3]
[6.6 3.  4.4 1.4]
[6.8 2.8 4.8 1.4]
[6.7 3.  5.  1.7]
[6.  2.9 4.5 1.5]
[5.7 2.6 3.5 1. ]
[5.5 2.4 3.8 1.1]
[5.5 2.4 3.7 1. ]
[5.8 2.7 3.9 1.2]
[6.  2.7 5.1 1.6]
[5.4 3.  4.5 1.5]
[6.  3.4 4.5 1.6]
[6.7 3.1 4.7 1.5]
[6.3 2.3 4.4 1.3]
```

```
[5.6 3.  4.1 1.3]
[5.5 2.5 4.  1.3]
[5.5 2.6 4.4 1.2]
[6.1 3.  4.6 1.4]
[5.8 2.6 4.  1.2]
[5.  2.3 3.3 1. ]
[5.6 2.7 4.2 1.3]
[5.7 3.  4.2 1.2]
[5.7 2.9 4.2 1.3]
[6.2 2.9 4.3 1.3]
[5.1 2.5 3.  1.1]
[5.7 2.8 4.1 1.3]
[6.3 3.3 6.  2.5]
[5.8 2.7 5.1 1.9]
[7.1 3.  5.9 2.1]
[6.3 2.9 5.6 1.8]
[6.5 3.  5.8 2.2]
[7.6 3.  6.6 2.1]
[4.9 2.5 4.5 1.7]
[7.3 2.9 6.3 1.8]
[6.7 2.5 5.8 1.8]
[7.2 3.6 6.1 2.5]
[6.5 3.2 5.1 2. ]
[6.4 2.7 5.3 1.9]
[6.8 3.  5.5 2.1]
[5.7 2.5 5.  2. ]
[5.8 2.8 5.1 2.4]
[6.4 3.2 5.3 2.3]
[6.5 3.  5.5 1.8]
[7.7 3.8 6.7 2.2]
[7.7 2.6 6.9 2.3]
[6.  2.2 5.  1.5]
[6.9 3.2 5.7 2.3]
[5.6 2.8 4.9 2. ]
[7.7 2.8 6.7 2. ]
[6.3 2.7 4.9 1.8]
[6.7 3.3 5.7 2.1]
[7.2 3.2 6.  1.8]
[6.2 2.8 4.8 1.8]
[6.1 3.  4.9 1.8]
[6.4 2.8 5.6 2.1]
[7.2 3.  5.8 1.6]
[7.4 2.8 6.1 1.9]
[7.9 3.8 6.4 2. ]
[6.4 2.8 5.6 2.2]
[6.3 2.8 5.1 1.5]
[6.1 2.6 5.6 1.4]
[7.7 3.  6.1 2.3]
```

```
 [6.3 3.4 5.6 2.4]
 [6.4 3.1 5.5 1.8]
 [6.  3.  4.8 1.8]
 [6.9 3.1 5.4 2.1]
 [6.7 3.1 5.6 2.4]
 [6.9 3.1 5.1 2.3]
 [5.8 2.7 5.1 1.9]
 [6.8 3.2 5.9 2.3]
 [6.7 3.3 5.7 2.5]
 [6.7 3.  5.2 2.3]
 [6.3 2.5 5.  1.9]
 [6.5 3.  5.2 2. ]
 [6.2 3.4 5.4 2.3]
 [5.9 3.  5.1 1.8]]

#Iris data Ground Truth result
print(iris.target)

print(iris.DESCR)

.. _iris_dataset:

Iris plants dataset
--------------------

**Data Set Characteristics:**

:Number of Instances: 150 (50 in each of three classes)
:Number of Attributes: 4 numeric, predictive attributes and the class
:Attribute Information:
    - sepal length in cm
    - sepal width in cm
    - petal length in cm
    - petal width in cm
    - class:
            - Iris-Setosa
            - Iris-Versicolour
            - Iris-Virginica

:Summary Statistics:

============== ==== ==== ======= ===== ====================
                Min  Max   Mean    SD   Class Correlation
============== ==== ==== ======= ===== ====================
sepal length:   4.3  7.9   5.84   0.83    0.7826
sepal width:    2.0  4.4   3.05   0.43   -0.4194
petal length:   1.0  6.9   3.76   1.76    0.9490  (high!)
petal width:    0.1  2.5   1.20   0.76    0.9565  (high!)
============== ==== ==== ======= ===== ====================
```

:Missing Attribute Values: None
:Class Distribution: 33.3% for each of 3 classes.
:Creator: R.A. Fisher
:Donor: Michael Marshall (MARSHALL%PLU@io.arc.nasa.gov)
:Date: July, 1988

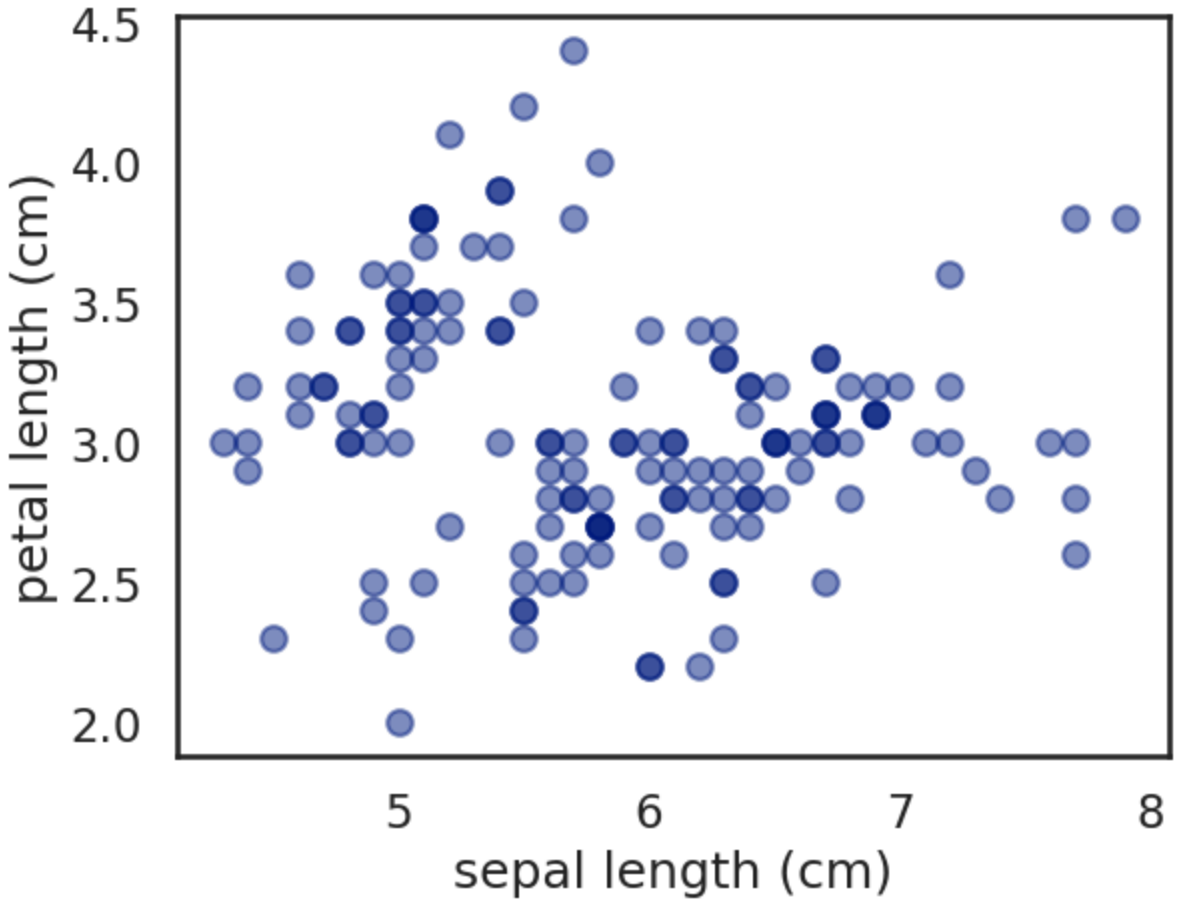
The famous Iris database, first used by Sir R.A. Fisher. The dataset is taken from Fisher's paper. Note that it's the same as in R, but not as in the UCI Machine Learning Repository, which has two wrong data points.

This is perhaps the best known database to be found in the pattern recognition literature. Fisher's paper is a classic in the field and is referenced frequently to this day. (See Duda & Hart, for example.) The data set contains 3 classes of 50 instances each, where each class refers to a type of iris plant. One class is linearly separable from the other 2; the latter are NOT linearly separable from each other.

.. dropdown:: References

- Fisher, R.A. "The use of multiple measurements in taxonomic problems" Annual Eugenics, 7, Part II, 179-188 (1936); also in "Contributions to Mathematical Statistics" (John Wiley, NY, 1950).
- Duda, R.O., & Hart, P.E. (1973) Pattern Classification and Scene Analysis (Q327.D83) John Wiley & Sons. ISBN 0-471-22361-1. See page 218.
- Dasarathy, B.V. (1980) "Nosing Around the Neighborhood: A New System Structure and Classification Rule for Recognition in Partially Exposed Environments". IEEE Transactions on Pattern Analysis and Machine Intelligence, Vol. PAMI-2, No. 1, 67-71.
- Gates, G.W. (1972) "The Reduced Nearest Neighbor Rule". IEEE Transactions on Information Theory, May 1972, 431-433.
- See also: 1988 MLC Proceedings, 54-64. Cheeseman et al"s AUTOCLASS II conceptual clustering system finds 3 classes in the data.
- Many, many more ...


```
samples = iris.data
x = samples[:,0]
y = samples[:,1]

sepal_length_width = np.array(list(zip(x, y)))
plt.scatter(x, y, alpha=0.5)

plt.xlabel('sepal length (cm)')
plt.ylabel('petal length (cm)')

plt.show()
```



png

Clustering using k-means from scikit-learn library

```
k=3
model = KMeans(n_clusters = k)

#use .fit() method to fit the model to the samples
model.fit(samples)
```

```
KMeans
KMeans(n_clusters=3)
```

```
# Use .predict() to determine the labels of samples
labels = model.predict(samples)
# Print the labels
print(labels)
```

```
[0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
 0 0 0 0 0 0 0 0 0 0 0 0 0 2 1 2 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1
 1 1 1 2 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 2 1 2 2 2 2 1 2 2 2 2
 2 2 1 1 2 2 2 2 1 2 1 2 1 2 2 1 1 2 2 2 2 2 1 2 2 2 2 1 2 2 2 1 2 2 2 1 2
 2 1]
```

Train-test splits

```
data = df.copy()
data.head()
```

	sepal_length	sepal_width	petal_length	petal_width	species
Id					
1	5.1	3.5	1.4	0.2	Iris-setosa
2	4.9	3.0	1.4	0.2	Iris-setosa
3	4.7	3.2	1.3	0.2	Iris-setosa
4	4.6	3.1	1.5	0.2	Iris-setosa
5	5.0	3.6	1.4	0.2	Iris-setosa

```
<div class="colab-df-buttons">


<button class="colab-df-convert" onclick="convertToInteractive('df-d24f0c84-6
        title="Convert this dataframe to an interactive table."
        style="display:none;">
```

```
</div>

from sklearn.model_selection import train_test_split

target = 'species'
X = data[['sepal_length', 'sepal_width', 'petal_length', 'petal_width']]
y = data[target]
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size = 0.2, ran
# it shows 80% of data is split for training and 20% of the data goes to testi
print(X_train.shape)
print(X_test.shape)

(120, 4)
(30, 4)
```

Logistic Regression

Logistic Regression is similar to Linear Regression but the target variable is categorical. It predicts the probability of an observation belonging to a category of the target variable. Sigmoid Function = ey / (1+ey) where ey = [co-efficients-matrix][feature-vector]

Source- https://www.google.com/ url?sa=i&url=http%3A%2F%2Fjuangabrielgomila.com%2Fen%2Flogistic-regression-derivation%2F&psig=AOvVaw0NnEbsC6SfeEZ5QqseuWbG&ust=1617370255285000&sou
T3e8CFQAAAAAdAAAAABAD

Import the necessary libraries from sciki-learn package

```
from sklearn.linear_model import LogisticRegression
from sklearn import metrics
lr = LogisticRegression()
lr.fit(X_train, y_train)
y_pred = lr.predict(X_test)
print('Test Accuracy for Scikit-Learn model:', metrics.accuracy_score(y_test,

Test Accuracy for Scikit-Learn model: 93.33333333333333 %
```

Evaluation Metrics

1. Jaccard-index
2. Confusion matrix
3. Logloss
4. F1-Score

```
from sklearn.metrics import classification_report
print(classification_report(y_test, y_pred))
```

```
                 precision    recall  f1-score   support

    Iris-setosa       1.00      1.00      1.00         9
Iris-versicolor       0.83      1.00      0.91        10
 Iris-virginica       1.00      0.82      0.90        11

       accuracy                           0.93        30
      macro avg       0.94      0.94      0.94        30
   weighted avg       0.94      0.93      0.93        30
```

About Dataset

Context

The two datasets are related to red and white variants of the Portuguese “Vinho Verde” wine. For more details, refer to [Cortez et al., 2009]. Due to privacy and logistic issues, only physicochemical (inputs) and sensory (the output) variables are available (e.g. there is no data about grape types, wine brand, wine selling price, etc.).

These datasets can be viewed as classification or regression tasks. The classes are ordered and not balanced (e.g. there are many more normal wines than excellent or poor ones). Outlier detection algorithms could be used to detect the few excellent or poor wines. Also, we are not sure if all input variables are relevant. So it could be interesting to test feature selection methods.

Content For more information, read [Cortez et al., 2009]. Input variables (based on physicochemical tests): 1 - fixed acidity 2 - volatile acidity 3 - citric acid 4 - residual sugar 5 - chlorides 6 - free sulfur dioxide 7 - total sulfur dioxide 8 - density 9 - pH 10 - sulphates 11 - alcohol Output variable (based on sensory data): 12 - quality (score between 0 and 10)

Acknowledgements This dataset is also available from the UCI machine learning repository, https://archive.ics.uci.edu/ml/datasets/wine+quality, to get both the dataset i.e. red and white vinho verde wine samples, from the north of Portugal, please visit the above link.

Please include this citation if you plan to use this database:

P. Cortez, A. Cerdeira, F. Almeida, T. Matos and J. Reis. Modeling wine preferences by data mining from physicochemical properties. In Decision Support Systems, Elsevier, 47(4):547-553, 2009.

Inspiration We kagglers can apply several machine-learning algorithms to determine which physiochemical properties make a wine ‘good’!

Relevant papers P. Cortez, A. Cerdeira, F. Almeida, T. Matos and J. Reis. Modeling wine preferences by data mining from physicochemical properties. In Decision Support Systems, Elsevier, 47(4):547-553, 2009.

```
import pandas as pd

test = pd.read_excel

df_wine = pd.read_csv('/winequality-white.csv', sep=';')
display(df_wine.head())
```

	fixed acidity	volatile acidity	citric acid	residual sugar	chlorides	free sulfur dioxide	total sulfur dioxide	density	pH	sulphates
0	7.0	0.27	0.36	20.7	0.045	45.0	170.0	1.0010	3.00	0.45
1	6.3	0.30	0.34	1.6	0.049	14.0	132.0	0.9940	3.30	0.49
2	8.1	0.28	0.40	6.9	0.050	30.0	97.0	0.9951	3.26	0.44
3	7.2	0.23	0.32	8.5	0.058	47.0	186.0	0.9956	3.19	0.40
4	7.2	0.23	0.32	8.5	0.058	47.0	186.0	0.9956	3.19	0.40

```
<div class="colab-df-buttons">


<button class="colab-df-convert" onclick="convertToInteractive('df-7de22379-1
        title="Convert this dataframe to an interactive table."
        style="display:none;">
```

```
</div>
```

Non-Graphical EDA for Wine Quality Dataset

Non-graphical EDA involves using statistical methods and data aggregations to understand the dataset. For the `df_wine` dataset, we can perform several non-graphical analyses to understand its structure, central tendencies, spread, and the distribution of its variables.

1. Descriptive Statistics: We can calculate summary statistics for all numerical columns, providing insights into their mean, median, standard deviation, quartiles, and range.

```
print('--- Descriptive Statistics for Wine Quality Dataset ---')
display(df_wine.describe())
```

```
--- Descriptive Statistics for Wine Quality Dataset ---
```

	fixed acidity	volatile acidity	citric acid	residual sugar	chlorides	free sulf dioxi
count	4898.000000	4898.000000	4898.000000	4898.000000	4898.000000	4898.0000
mean	6.854788	0.278241	0.334192	6.391415	0.045772	35.308085
std	0.843868	0.100795	0.121020	5.072058	0.021848	17.007137
min	3.800000	0.080000	0.000000	0.600000	0.009000	2.000000
25%	6.300000	0.210000	0.270000	1.700000	0.036000	23.000000
50%	6.800000	0.260000	0.320000	5.200000	0.043000	34.000000
75%	7.300000	0.320000	0.390000	9.900000	0.050000	46.000000
max	14.200000	1.100000	1.660000	65.800000	0.346000	289.00000

```
<div class="colab-df-buttons">


<button class="colab-df-convert" onclick="convertToInteractive('df-d3f623c7-f
        title="Convert this dataframe to an interactive table."
```

```
        style="display:none;">
```

```
</div>
```

2. Value Counts for Categorical/Ordinal Data: For the ‘quality’ column, which is an ordinal variable, checking value counts helps us understand the distribution of wine quality ratings.

```
print('\n--- Value Counts for Wine Quality (Ordinal Variable) ---')
display(df_wine['quality'].value_counts().sort_index())
```

```
--- Value Counts for Wine Quality (Ordinal Variable) ---
```

	count
quality	
3	20
4	163
5	1457
6	2198
7	880
8	175
9	5

dtype: int64

Explanation of Answer:

By examining these non-graphical outputs, we can immediately glean valuable insights:

- **Descriptive Statistics:** For instance, `df_wine.describe()` shows the average `fixed acidity`, `pH`, `alcohol` content, and their variability. It also reveals the minimum and maximum values for each physicochemical property, helping to identify potential outliers or data ranges. We can see that ‘quality’ ranges from 3 to 9, with an average around 5.8.

- **Value Counts for Quality:** `df_wine['quality'].value_counts()` directly shows how many wines fall into each quality rating (e.g., how many wines are rated ‘6’, how many are ‘5’, etc.). This immediately highlights if the dataset is balanced across different quality ratings or if certain ratings are more common. In this case, ‘quality’ ratings of 6, 5, and 7 are the most frequent.

Q1 **Describe all three data set in terms of following : how many features, their categories (num,text,categorical, nominal,ordinal etc), number of tuples, missing values.**

```
print('--- Wine Quality Dataset Info ---')
df_wine.info()
print('\n--- Wine Quality Dataset Missing Values ---')
df_wine.isnull().sum()
```

```
--- Wine Quality Dataset Info ---
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 4898 entries, 0 to 4897
Data columns (total 12 columns):
 #   Column                Non-Null Count  Dtype
---  ------                --------------  -----
 0   fixed acidity         4898 non-null   float64
 1   volatile acidity      4898 non-null   float64
 2   citric acid           4898 non-null   float64
 3   residual sugar        4898 non-null   float64
 4   chlorides             4898 non-null   float64
 5   free sulfur dioxide   4898 non-null   float64
 6   total sulfur dioxide  4898 non-null   float64
 7   density               4898 non-null   float64
 8   pH                    4898 non-null   float64
 9   sulphates             4898 non-null   float64
 10  alcohol               4898 non-null   float64
 11  quality               4898 non-null   int64
dtypes: float64(11), int64(1)
memory usage: 459.3 KB
```

```
--- Wine Quality Dataset Missing Values ---
```

	0
fixed acidity	0
volatile acidity	0
citric acid	0
residual sugar	0
chlorides	0
free sulfur dioxide	0
total sulfur dioxide	0
density	0
pH	0
sulphates	0
alcohol	0
quality	0

dtype: int64

Dataset Descriptions

Here's a description of each dataset based on the current notebook state:

1. Iris Dataset (`df`): * **Number of Features:** 5 * **Features and Categories:** * `sepal_length` (Numerical, float64) * `sepal_width` (Numerical, float64) * `petal_length` (Numerical, float64) * `petal_width` (Numerical, float64) * `species` (Categorical/Nominal, object) * **Number of Tuples:** 150 * **Missing Values:** None

2. California Housing Dataset (loaded as `california_data`, described as 'Air Quality Dataset' in markdown cells): * **Number of Features:** 9 * **Features and Categories:** * `longitude` (Numerical, float64) * `latitude` (Numerical, float64) * `housing_median_age` (Numerical, float64) * `total_rooms` (Numerical, float64) * `total_bedrooms` (Numerical, float64) * `population` (Numerical, float64) * `households` (Numerical, float64) * `median_income` (Numerical, float64) * `median_house_value` (Numerical, float64) * **Number of Tuples:** 17,000 * **Missing Values:** None

3. Wine Quality Dataset (`df_wine`): * **Number of Features:** 12 * **Features and Categories:** * `fixed acidity` (Numerical, float64) * `volatile acidity` (Numerical, float64) * `citric acid` (Numerical, float64) * `residual sugar` (Numerical, float64) * `chlorides` (Numerical, float64) * `free sulfur dioxide` (Numerical, float64) * `total sulfur dioxide` (Numerical, float64) * `density` (Numerical, float64) * `pH` (Numerical, float64) * `sulphates` (Numerical, float64) * `alcohol` (Numerical, float64) * `quality` (Numerical/Ordinal, int64) - This is the

target variable, an ordinal score. * **Number of Tuples:** 4,898 * **Missing Values:** None

```
from google.colab import drive
drive.mount('/content/drive')
```

```
Drive already mounted at /content/drive; to attempt to forcibly remount, call
```

Q2 **In IRIS data set which non graphical EDA is performed ?**

Ans. For the Iris dataset, the following non-graphical Exploratory Data Analysis (EDA) techniques were performed:

Displaying the head of the DataFrame (df.head()): This provides a quick overview of the data structure and initial values. Calculating descriptive statistics (df.describe()): This generated summary statistics like count, mean, standard deviation, min, max, and quartiles for numerical columns. Calculating median values grouped by species (df.groupby('species').median()): This provided insight into the central tendency of features for each Iris species. Creating a custom statistics DataFrame (new_df): This involved calculating ranges and selecting specific descriptive statistics (mean, range, 25%, 50% (median), 75%) for further examination. Performing multiple aggregations by species (df.groupby('species').agg(['mean', 'median']) and df.groupby('species').agg(agg_dict)): These operations calculated various statistics (mean, median, max) for different features, grouped by the 'species' column, to understand differences between the groups. Checking for missing values (implied by the dataset description in cell f5682662): Although not explicitly shown with df.isnull().sum() for df, this is a standard non-graphical EDA step to ensure data completeness, and the description confirms no missing values.

Q3 **In Air Quality data set which graph is used to perform multivariate analysis?**

Ans. In the notebook's 'Air Quality data set' section, which uses the California Housing dataset, the following graphs are used to perform multivariate analysis:

Correlation Matrix Heatmap (Cell QzTjEpqSv4fc): This heatmap visualizes the pairwise correlations between median_house_value, median_income, total_rooms, and total_bedrooms. By showing the relationships among multiple numerical variables simultaneously, it provides a multivariate view of their associations.

Pairplot (Cell x-ntrwQcv4fd): This plot generates a grid of scatter plots for each pair of variables (median_house_value, median_income, total_rooms, total_bedrooms), and histograms (or KDEs) for each individual variable on the diagonal. This is a powerful tool for understanding the relationships and distributions of multiple variables in a dataset.

Q4**In Wine quality data set can you do non graphical EDA ? Explain your answer.** Ans. es, absolutely! Non-graphical Exploratory Data Analysis is very

useful for the Wine Quality dataset. It helps us understand the data's characteristics without relying on visualizations.

I'll explain how, and then generate code to perform some key non-graphical EDA steps on the df_wine DataFrame, such as obtaining descriptive statistics and checking the distribution of the 'quality' variable.

The non-graphical EDA for the Wine Quality dataset (df_wine) has been successfully executed, providing some valuable insights:

Descriptive Statistics: The df_wine.describe() output shows the average fixed acidity (around 6.85), pH (around 3.19), and alcohol content (around 10.51), along with their variability (standard deviation), and ranges (min/max). We can also see that the quality ratings range from 3 to 9, with an average of approximately 5.88.

Value Counts for Quality: The df_wine['quality'].value_counts() reveals the distribution of wine quality ratings. The most frequent quality ratings are 6 (2198 wines), 5 (1457 wines), and 7 (880 wines). This indicates that the dataset is skewed towards wines of moderate quality, with very few wines rated as 3 or 9. This kind of imbalance can be important for modeling tasks.

I have successfully performed and explained the non-graphical EDA for the Wine Quality dataset. We observed descriptive statistics for all numerical features, including measures like mean, standard deviation, and min/max values. We also examined the value counts for the 'quality' variable, which revealed an imbalanced distribution, with most wines rated 5, 6, or 7.